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Stable Deep Reinforcement Learning via Isotropic Gaussian Representations

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Abstract

Deep reinforcement learning systems often suffer from unstable training dynamics due to non-stationarity, where learning objectives and data distributions evolve over time. We show that under non-stationary targets, isotropic Gaussian embeddings are provably advantageous. In particular, they induce stable tracking of time-varying targets for linear readouts, achieve maximal entropy under a fixed variance budget, and encourage a balanced use of all representational dimensions–all of which enable agents to be more adaptive and stable. Building on this insight, we propose the use of Sketched Isotropic Gaussian Regularization for shaping representations toward an isotropic Gaussian distribution during training. We demonstrate empirically, over a variety of domains, that this simple and computationally inexpensive method improves performance under non-stationarity while reducing representation collapse, neuron dormancy, and training instability. **Our code is available at `IsoGaussian-DRL`.**

1. Introduction

Deep reinforcement learning (deep RL) has achieved striking successes across a wide range of domains, including robotics, control, game playing, and large-scale decision-making (Vinyals et al., 2019; Bellemare et al., 2020; Fawzi et al., 2022; Schwarzer et al., 2023). Despite this progress, optimization pathologies such as instability, plasticity loss (Nikishin et al., 2022), neuron dormancy (Sokar et al., 2023), and representation collapse (Moalla et al., 2024; Mayor et al., 2025) remain persistent and often limit both performance and scalability. A defining challenge of deep RL is its inherently non-stationary nature. Unlike supervised or standard self-supervised learning, the data distribution, learning targets, and optimization landscape evolve continually as the agent updates its policy and interacts with the environment (Sutton & Barto, 1988; van Hasselt et al., 2018). This non-stationarity has been repeatedly linked to degraded representation quality, unstable gradients, and brittle learning dynamics, ultimately reducing effective capacity over long training horizons (Kumar et al., 2021; Obando-Ceron et al., 2025; Liu et al., 2025c; Castanyer et al., 2025). Most prior work addressing instability in deep RL focuses on algorithmic or architectural interventions, including modified update rules (Hessel et al., 2018), auxiliary objectives (Castro et al., 2021; Schwarzer et al., 2021), target networks, or carefully designed architectural heuristics (Ceron et al., 2024c;b; Lee et al., 2025; Castanyer et al., 2025). While often effective, these approaches typically address downstream symptoms of instability and may introduce additional complexity or tuning requirements. In contrast, the role of representation geometry, specifically, how the statistical structure of learned representations interacts with non-stationary targets, has received comparatively limited attention in deep RL.

Recent advances in self-supervised learning suggest that representation geometry plays a central role in stability and generalization. In particular, LeJEPA (Balestriero & LeCun, 2025) demonstrates that enforcing simple statistical structure, namely isotropy and approximate Gaussianity, yields representations that generalize robustly across diverse downstream tasks. This setting closely mirrors deep RL, where the effective task and target distribution evolve over time and are not known a priori. Motivated by this parallel, we revisit instability in deep RL through the lens of representation geometry and ask: *what properties should representations satisfy to remain stable under continuously evolving targets?*

We show that isotropic Gaussian representations are particularly well suited to this regime. For linear readouts tracking non-stationary targets, such representations minimize sensitivity to drift, maximize entropy under a fixed variance budget, and promote balanced utilization of representational dimensions. Together, these properties mitigate feature collapse, reduce neuron dormancy, and stabilize learning dynamics under distributional shift. These guarantees highlight isotropic Gaussian structure as a principled target for representations in non-stationary learning systems.

To empirically study this hypothesis, we evaluate meth-

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ods that encourage isotropic Gaussian structure in learned representations within standard deep RL pipelines. Our experiments span discrete and continuous control benchmarks, including the Arcade Learning Environment (Bellemare et al., 2013) and Isaac Gym (Makoviychuk et al., 2021a), and cover both value-based and policy-gradient methods such as PQN (Gallici et al., 2025) and PPO (Schulman et al., 2017). Across settings, we observe consistent improvements in training stability, representation quality, and performance under non-stationarity. These results suggest that representation geometry is central to stabilizing learning under non-stationarity such as deep RL.

2. Preliminaries

Deep Reinforcement Learning. Deep RL addresses sequential decision-making problems in which an agent learns a policy mapping states to actions through interaction with an environment, typically using neural networks as function approximators (Mnih et al., 2015). This interaction is formalized as a Markov decision process (MDP) $(\mathcal{S}, \mathcal{A}, P, r, \gamma)$ (Puterman, 1994). A policy $\pi_\theta(a \mid s)$ is optimized via gradient-based methods, inducing a discounted state–action occupancy measure $d_{\pi_\theta}(s, a)$. As policy parameters are updated, both the policy and the induced data distribution evolve, resulting in a continuously shifting training distribution even under fixed environment dynamics. Most deep RL algorithms rely on bootstrapped learning targets. Value-based and actor–critic methods, including DQN (Mnih et al., 2015), PQN (Gallici et al., 2025), SAC (Haarnoja et al., 2018), and TD3 (Fujimoto et al., 2018), optimize objectives of the form $\mathcal{L}_{\text{TD}}(\phi) = \mathbb{E}_{(s,a,r,s')\sim d_{\pi_\theta}}\left[(Q_\phi(s,a) - y_{\theta,\phi}(s,a,r,s'))^2\right]$; where the target $y_{\theta,\phi}$ depends on learned quantities. As a result, both the data distribution and learning targets evolve throughout training, introducing inherent non-stationarity and contributing to optimization instability (Vincent et al., 2025; Hendawy et al., 2025). Empirically, this non-stationarity leads to representation drift, neuron dormancy, and loss of effective capacity, often manifested as rank collapse and degraded adaptability (Lyle et al., 2022; Nikishin et al., 2022; Moalla et al., 2024; Tang et al., 2025).

Representation Learning in Deep RL. Deep RL agents typically decompose their networks into a shared representation $h_\eta(s) \in \mathbb{R}^d$ and task-specific prediction heads, e.g., $Q_\phi(s, a) = g_\omega(h_\eta(s), a)$ (Fujimoto et al., 2023; Echchahed & Castro, 2025). Although objectives are defined over scalar signals such as returns or advantages, gradients propagate through the prediction head into the representation. Consequently, deep RL simultaneously performs control and representation learning under indirect, noisy, and non-stationary supervision (Igl et al., 2021). Sampling $s \sim d_{\pi_\theta}$ induces a distribution over representations $z = h_\eta(s)$, whose geometry evolves throughout training.

3. Isotropic Gaussian Representations Promote Stability Under Non-Stationarity

We study the role of isotropic Gaussian representations by analyzing learning dynamics under drifting targets. We show that when embeddings are constrained to be isotropic and Gaussian, the resulting optimization dynamics has a stable fixed point, and the tracking error remains bounded and decreases over time. This provides a theoretical explanation for the empirical robustness of isotropic Gaussian representations under non-stationary training and motivates explicitly enforcing this type of representation in deep RL. Our analysis is based on the formal model in App. B, where representation learning is viewed as a tracking problem under drifting supervision. This analysis is inspired by Lemmas 1 and 2 in (Balestriero & LeCun, 2025), which show that, without knowledge of the downstream task, an isotropic representation is optimal for learning a wide range of downstream tasks. Building on this insight, we model non-stationarity as learning a sequence of tasks that evolve over time, and we seek conditions that ensure task changes induced by non-stationarity remain learnable.

3.1. Linear Tracking Under Non-Stationary Targets

We consider a linear critic on top of a learned embedding, $Q_\theta(s, a) = w^\top \phi(s, a)$, where $\phi(s, a) \in \mathbb{R}^d$ is the penultimate-layer embedding and $w \in \mathbb{R}^d$ is the last-layer weight vector. The TD target is time-varying, $y_t = r + \gamma Q_{\theta^-}(s', a')$, which induces non-stationarity in the learning problem. To characterize the resulting dynamics, define the second-order statistics $\Sigma_\phi(t) := \mathbb{E}[\phi\phi^\top]$, $b_t := \mathbb{E}[\phi y_t]$, and the expected critic loss as $\mathcal{L}_t(w) = w^\top \Sigma_\phi(t) w - 2w^\top b_t + \mathbb{E}[y_t^2]$ (obtained by taking expectation over critic loss shown in Eq. 6).

Under continuous-time gradient flow, the dynamics are given by $\dot{w}(t) = -\nabla_w \mathcal{L}_t(w) = -2\Sigma_\phi(t)w + 2b_t$. At each time t, the instantaneous minimizer is $w_t^* = \Sigma_\phi(t)^{-1} b_t$, which varies over time due to non-stationarity in the targets, policy, and representation distribution. We define the tracking error as $e(t) := w(t) - w_t^*$, and study stability using the Lyapunov function $\Gamma := \|e(t)\|_2^2$.

Theorem 3.1 (Tracking error dynamics)**.** *Assume that $\Sigma_\phi(t) \succ 0$ is constant over time (e.g., enforced by regularization) and that b_t is differentiable. Under gradient flow, the time derivative of Γ satisfies*

$$\dot{\Gamma} = -4\, e(t)^\top \Sigma_\phi(t)\, e(t) \;-\; 2\, e(t)^\top \Sigma_\phi(t)^{-1} \dot{b}_t. \quad (1)$$

Proof sketch. Writing $\dot{e}(t) = \dot{w} - \dot{w}_t^*$ and using $\dot{w} = -2\Sigma_\phi w + 2b_t$ together with $w_t^* = \Sigma_\phi^{-1} b_t$ (and fixed Σ_ϕ), we obtain $\dot{e}(t) = -2\Sigma_\phi e(t) - \Sigma_\phi^{-1}\dot{b}_t$. The claim follows from $\dot{\Gamma} = 2e(t)^\top \dot{e}(t)$. See Appendix B for the full proof. □

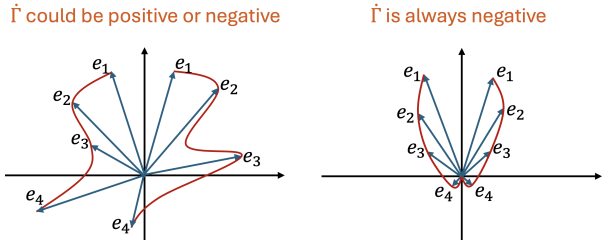

Figure 1: **Illustration of two tracking regimes. Left:** the norm of the tracking error exhibits non-monotonic behavior and fails to converge, indicating unstable tracking. **Right:** the tracking error decreases monotonically and converges to zero, corresponding to stable tracking dynamics.

Geometric Implications for Stability. The decomposition in Eq. 1 highlights competing effects governing stability under non-stationary targets. The first term (in blue) is a strict contraction whose strength depends on the spectrum of Σ_ϕ, while the second term (in red) captures drift induced by target non-stationarity through $\dot{b}_t$. The geometry of the learned representation plays a central role in determining whether the tracking error decays or grows over time.

1. **Tracking-error contraction.** Stability requires $\dot{\Gamma} < 0$ for all error directions, ensuring monotonic decay of the tracking error (see Fig. 1). While the contraction term $-4\, e^\top \Sigma_\phi e$ is always negative, its magnitude depends on the conditioning of Σ_ϕ and can vary significantly across directions.
2. **Effect of isotropy under non-stationarity.** The drift term $-2\, e^\top \Sigma_\phi^{-1} \dot{b}_t$ is amplified along poorly conditioned directions. Under a fixed total variance budget, anisotropic representations increase the likelihood that some directions exhibit weak contraction and large drift. Enforcing isotropy equalizes contraction across directions and reduces the probability that $\dot{\Gamma}$ becomes positive.
3. **Gaussian representations and tail behavior.** Among distributions with fixed second-order moments, Gaussian distributions maximize entropy and exhibit tail decay. This limits the occurrence of rare but harmful error directions that could dominate the drift term, while preserving maximal information content in the representation.

3.2. Why Isotropy Stabilizes Tracking Dynamics

To ensure that $\dot{\Gamma}$ is negative for all tracking error vectors, the contraction induced by the first term in Eq. 1 must dominate the second term, whose sign may vary. The strength of this contraction is governed by the geometry of the covariance matrix Σ_ϕ. Restricting the tracking error to the unit sphere, the weakest contraction occurs along the eigenvector corresponding to the smallest eigenvalue of Σ_ϕ. Equalizing the contraction across all directions therefore requires distributing the total variance uniformly across dimensions, which implies that all eigenvalues of Σ_ϕ are equal. The second term in Eq. 1 can be either positive or negative, and large magnitudes may destabilize the dynamics. As shown in Appendix B.1.4, its absolute value is upper bounded by the condition number of Σ_ϕ. Minimizing this bound requires the smallest possible condition number, which is achieved precisely when the covariance matrix is isotropic.

3.3. Why Gaussianity Controls Drift Variance

Among all isotropic distributions, the Gaussian is particularly well suited for stabilizing the dynamics induced by Eq. 1. The second term in this equation can fluctuate in sign and may occasionally dominate the contraction term, leading to positive values of $\dot{\Gamma}$. As shown in Appendix B.1.5, a Gaussian distribution minimizes the variance of this term, thereby reducing the likelihood of large destabilizing deviations. Distributions with heavier tails introduce higher-order moment contributions that increase this risk. This effect can be understood through moment concentration. The embedding vectors enter the dynamics through $\dot{b}_t$, and for any random variable X, $\mathbb{P}(|X-\mu| \geq t) \leq \mathbb{E}[|X-\mu|^p]/t^p$.

Large higher-order moments therefore translate directly into rare but severe spikes in the magnitude of the second term. By Isserlis' theorem (Isserlis, 1918), all higher-order moments of a Gaussian distribution are fully determined by its covariance, preventing such uncontrolled fluctuations. From an information-theoretic perspective, a Gaussian distribution further maximizes entropy among all continuous distributions with a fixed covariance. As a result, it achieves the least-structured, most unbiased representation compatible with isotropy, balancing expressiveness with stability.

3.4. Sketched Isotropic Gaussian Regularization

In Sections 3.2 and 3.3, we argued that isotropy and Gaussian tail behavior are desirable properties for stabilizing learning under non-stationarity. This raises two practical questions: *(i)* how to measure deviations from an isotropic Gaussian representation distribution, and *(ii)* how to enforce such structure efficiently during training. To this end, we adopt *Sketched Isotropic Gaussian Regularization* (SIGReg) (Balestriero & LeCun, 2025), a lightweight regularizer recently introduced in self-supervised learning. SIGReg avoids directly matching high-dimensional distributions by projecting embeddings onto a small number of random directions and applying a univariate distribution-matching loss to each projection. By resampling directions across iterations, SIGReg enforces isotropy and Gaussianity in expectation while remaining computationally efficient.

Formally, given an embedding $\phi \in \mathbb{R}^d$ and random unit vec-

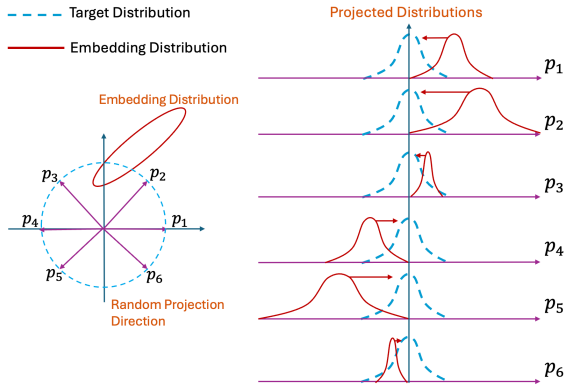

Figure 2: **Directly shaping a multivariate distribution.** SIGReg first projects the embeddings onto a small set of random directions (p_i: sketching), producing multiple univariate distributions. Each projection is then matched to the corresponding univariate target distribution.

tors $\{v_k\}_{k=1}^{K}$, SIGReg operates on the projected variables $z_k = v_k^\top \phi$. The regularization loss matches the empirical distribution of each z_k to a zero-mean Gaussian with variance σ^2 using a characteristic-function-based objective (Balestriero & LeCun, 2025) (see Fig. 2). This construction yields a fully differentiable regularizer that simultaneously controls isotropy and tail behavior. In our experiments, we study the impact of each component through targeted ablations.

Key Insight: Isotropy and Stability

When the feature covariance satisfies $\Sigma_\phi = \sigma^2 I$, the contraction term becomes isotropic and the tracking error is uniformly damped across all directions. In contrast, ill-conditioned representations induce anisotropic contraction, amplifying the effect of target drift through $\Sigma_\phi^{-1}\dot{b}_t$. As a consequence, non-isotropic embeddings can lead to unstable or directionally biased tracking behavior under non-stationarity.

4. Empirical Results

4.1. CIFAR-10 Under Distribution Shift

To isolate the effect of non-stationarity independently of control, exploration, and environment dynamics, we first study a controlled supervised setting based on CIFAR-10 (Krizhevsky et al., 2009). Following prior work (Igl et al., 2021; Sokar et al., 2023; Lyle et al., 2024; Castanyer et al., 2025; Obando-Ceron et al., 2025), we introduce non-stationary targets via a shuffled-label protocol, in which label assignments are periodically permuted during training. This induces a sequence of related but non-identical prediction problems, serving as a proxy for the drifting targets arising from bootstrapped updates in deep RL. This setup is widely used to study representation drift and neuron dormancy under non-stationarity, as it exposes similar representational stresses while avoiding confounding effects from policy learning and exploration (Sokar et al., 2023; Castanyer et al., 2025). Operating in a fully supervised regime allows us to directly analyze how representation geometry evolves under target drift. All experimental hyperparameters and training details are reported in App. E.

We compare standard training against models encouraged toward isotropic representations, keeping the architecture, optimizer, and learning rate fixed. We track classification performance, effective rank, and the fraction of dormant neurons under distribution shift (see Fig. 3)[1]. Under standard training, representations quickly become anisotropic, with variance concentrating along few directions, accompanied by rank collapse and reduced adaptability. By contrast, encouraging isotropy preserves balanced variance, reduces neuron dormancy, and leads to more stable performance and faster recovery after shifts. These effects closely mirror behaviors observed in deep RL under policy-induced distributional drift.

4.2. Deep Reinforcement Learning

We evaluate isotropic Gaussian representations in deep RL, where non-stationarity arises naturally from policy improvement and bootstrapped learning targets. We focus on online deep RL agents trained in non-stationary regimes, where the distribution of visited states evolves continuously as the policy improves. This setting is particularly sensitive to representation collapse, gradient interference, and plasticity loss, which can silently degrade effective capacity even when training appears numerically stable.

Setup. We evaluate the effect of enforcing isotropic Gaussian representations in Parallelized Q-Networks (PQN) (Gallici et al., 2025). Experiments are conducted on the Atari-10 subset (Aitchison et al., 2023) of the Arcade Learning Environment (ALE) (Bellemare et al., 2013), following standard evaluation protocols (Agarwal et al., 2021). For each experiment, we compare a baseline implementation against an identical version in which representations are regularized toward an isotropic Gaussian distribution. All deep RL agents use the same convolutional encoders, optimizer hyperparameters, and training schedules (see App. E). The SIGReg regularizer is applied only to the learned representations and does not modify the policy or value objectives.

We empirically analyze the effect of encouraging isotropic Gaussian representations via explicit minimization of the SIGReg loss. We observe consistent improvements in repre-

[1]Unless otherwise specified, results are averaged over 5 seeds.

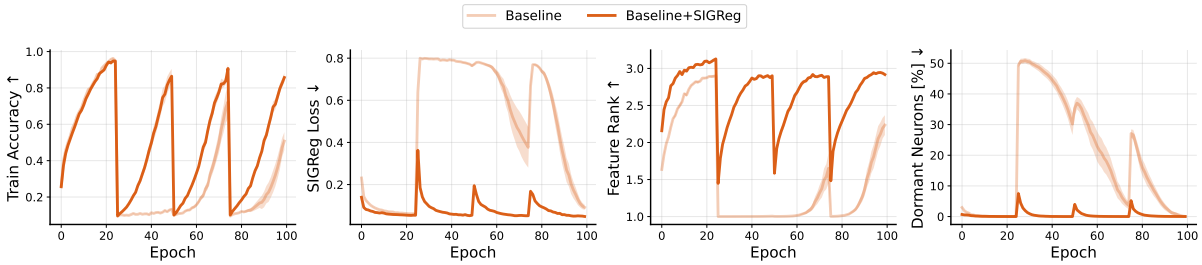

Figure 3: **Non-stationary CIFAR-10.** Training under repeated label shuffling. The baseline shows poor recovery after each shift, with SIGReg loss spikes, rank collapse, and increased dormancy. Enforcing isotropic Gaussian representations stabilizes training, accelerates recovery, preserves rank, and reduces dormancy.

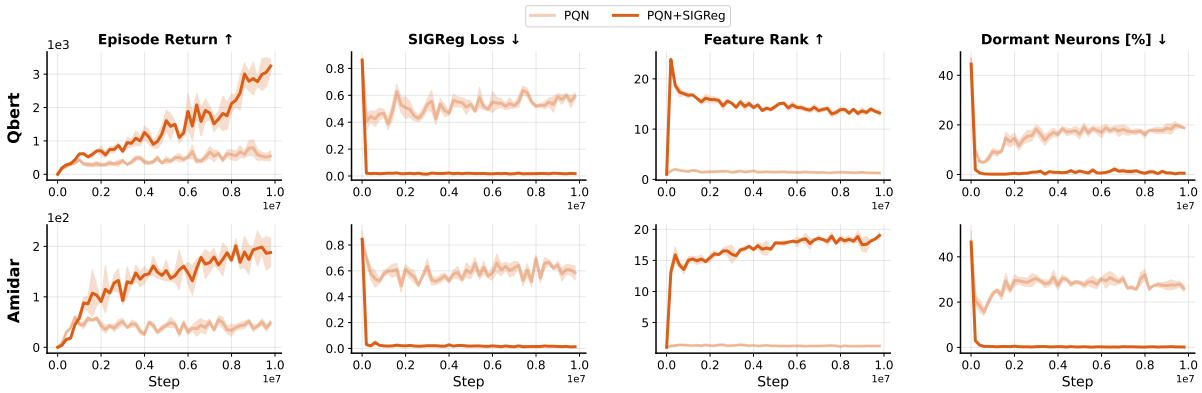

Figure 4: **Two Atari-10 games (PQN).** Without isotropy regularization, representations exhibit rank collapse, increased neuron dormancy, and early performance saturation. Encouraging isotropic geometry leads to improved representation quality and higher, more stable performance.

sentation quality, reduced neuron dormancy, and improved asymptotic performance. These results indicate that promoting isotropic Gaussian structure in representations mitigates common optimization pathologies in deep RL.

Effect on Representation Rank. Encouraging isotropic Gaussian structure in the learned representations leads to a systematic increase in embedding rank throughout training. This behavior is consistent with the objective of maintaining well-conditioned covariance structure, preventing collapse along low-variance directions. For PQN, the evolution of feature rank shows that representations remain high-rank over time, unlike the baseline, which exhibits progressive rank degradation. This indicates that enforcing isotropy stabilizes representation geometry under non-stationary training, preserving expressive capacity as learning progresses (see Figures 4 and 5).

Reduction of Neuron Dormancy. Alongside improved rank, isotropic Gaussian representations substantially reduce neuron dormancy in PQN. Maintaining high-entropy, well-spread representations prevents activations from concentrating on a small subset of units, thereby sustaining gradient flow across the network. As a result, fewer neurons become inactive over time, mitigating a common form of plasticity loss in deep RL. This suggests that representation geometry plays a direct role in preserving network capacity under continual bootstrapping (see Fig. 4).

Temporal Evolution of Representation Distributions. To better understand how isotropic Gaussian structure manifests during training, we analyze the temporal evolution of the learned representation distributions under PQN. Fig. 5 visualizes this evolution using a two-dimensional PCA projection of the embedding covariance at successive training stages. Without explicit constraints on representation geometry, the embeddings progressively collapse onto a small number of dominant principal directions, resulting in highly anisotropic distributions with most variance concentrated in a few components. In contrast, when representations are encouraged to follow an isotropic Gaussian structure, the projected covariance becomes approximately circular and centered at the origin. This indicates both isotropy and a more uniform allocation of variance across dimensions, corresponding to a higher effective dimensionality. These observations provide direct empirical evidence that isotropic Gaussian representations counteract representational collapse and maintain well-conditioned feature spaces throughout training.

Performance Implications in PQN. We next examine whether these representational improvements translate into

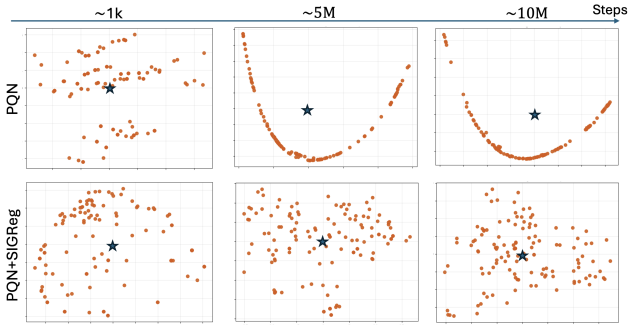

Figure 5: **2D PCA of embedding covariance over training.** Without constraints, representations collapse onto a few dominant principal components. Encouraging isotropic Gaussian structure yields more evenly distributed variance and higher effective dimensionality, reflected in reduced concentration on the leading components (PQN: $[0.4, 0.2] \rightarrow [0.9, 0.8]$ vs. PQN+SIGReg: $[0.3, 0.2] \rightarrow [0.1, 0.1]$).

gains in learning performance. Using Area Under the Curve (AUC) as our primary metric, which captures both learning speed and final performance, we observe consistent improvements over the baseline PQN agent. These gains indicate that stabilizing representation geometry through isotropic Gaussian structure not only improves internal metrics such as rank and dormancy, but also yields tangible benefits in control performance. Importantly, these improvements are achieved without introducing additional architectural complexity or second-order optimization, highlighting representation regularization as a lightweight and effective mechanism for stabilizing PQN training. Fig. 4 shows improved sample efficiency and final performance on two Atari games when promoting isotropic Gaussian representations. See Fig. 6 for additional results across more ALE games.

4.3. Analysis of Design Choices

In this section, we analyze how different design choices for promoting isotropic representations affect learning under PQN. Specifically, we study the impact of (i) alternative isotropic target distributions with heavier tails or imposing isotropy alone through covariance whitening, as proposed in VICReg (Bardes et al., 2021) in the self-supervised learning literature, and (ii) enforcing only symmetry (minimizing the imaginary part) or only tail behavior (minimizing the real part). These ablations allow us to disentangle which statistical properties of isotropic Gaussian representations are most critical for stable and efficient deep RL.

Alternative Isotropic Distributions. We first compare isotropic Gaussian representations with alternative isotropic distributions exhibiting heavier tails, namely Laplacian and Logistic distributions. While heavier-tailed distributions may be sufficient in some regimes, our results indicate that they are consistently less effective under PQN. As shown in Table 2, isotropic Gaussian representations yield the strongest and most reliable gains, improving approximately 90% of games with large average relative improvements. In contrast, Laplacian and Logistic distributions achieve smaller gains despite improving a similar fraction of environments. It has also been observed that achieving isotropy via covariance whitening is not enough (last row in Table 2). This suggests that, beyond isotropy, fast tail decay plays a central role in stabilizing learning and preventing extreme activations under non-stationary targets.

Role of Symmetry and Tail Decay. To further isolate the role of different statistical properties, we evaluate variants that enforce only symmetry (imaginary part) or only tail behavior (real part). Table 2 shows that neither property alone matches the performance of full isotropic Gaussian representations. Enforcing tail behavior is generally more effective than enforcing symmetry alone, likely because intrinsic regularization already induces approximate symmetry. However, jointly enforcing both consistently gives the strongest and most stable improvements, showing that the benefit comes from their combination rather than from either property in isolation.

4.4. Implicit Isotropy in Stabilization Methods

We further investigate the connection between isotropic Gaussian representations and strong stabilization mechanisms introduced by Castanyer et al. (2025), namely Kronecker-factored optimization and multi-skip residual architectures, both of which were designed to mitigate gradient pathologies and improve stability at scale. Fig. 9 reports results for Parallelized Q-Networks (PQN) for Atari-10 games. Across nearly all games, Kronecker-factored optimization consistently induces representations that are substantially closer to isotropic Gaussian distribution (lower SIGReg loss) than those learned with first-order methods such as RAdam. Importantly, this effect emerges without any explicit representation regularization. Similarly, multi-skip architectures, which were introduced to stabilize gradient propagation, also implicitly promote better-conditioned and more isotropic representations (see Fig. 10).

Although the Kronecker-factored optimizer can be effective and provide significant improvements, it requires additional memory and computation due to gradient curvature estimation. Therefore, it is desirable to achieve similar performance with first-order optimizers. Based on observations from Fig. 9, we hypothesize that part of the stability and performance gains provided by these optimizers are the effect of the way they shape the geometry of the learned representations. Therefore, if we explicitly shape the embeddings of a baseline model, we should expect to see a reduction in the performance gap. Consistent with this idea,

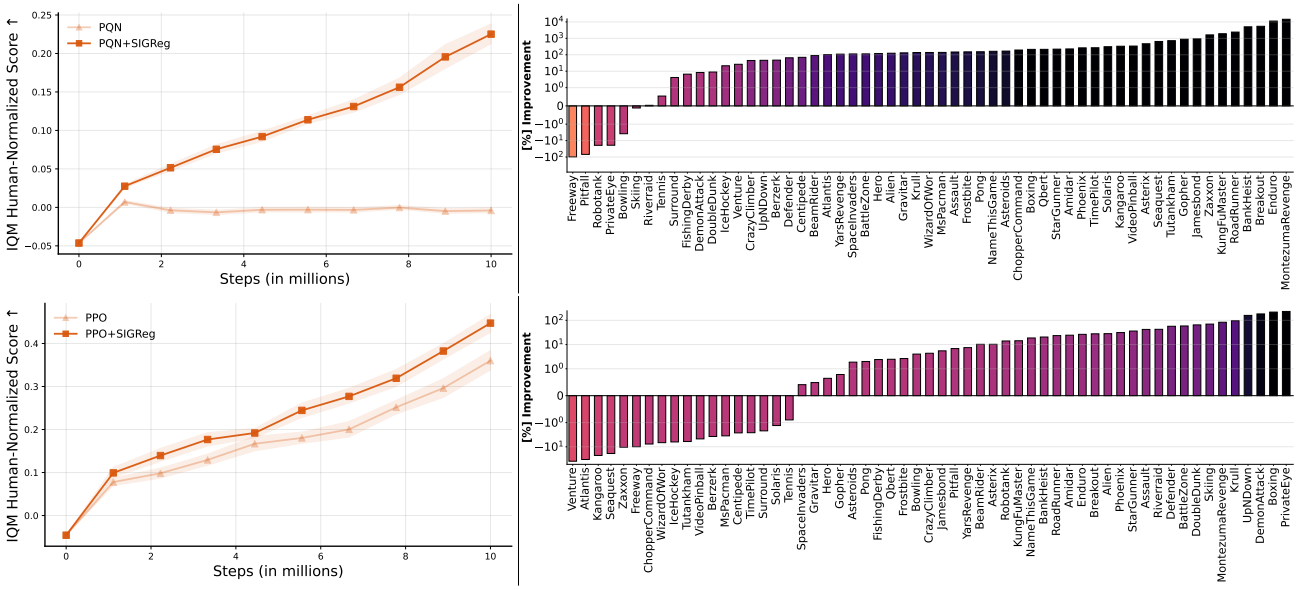

Figure 6: **Full Atari suite. Effect of isotropic Gaussian regularization. Left:** IQM human-normalized learning curves as a function of environment steps for PQN and PPO, with and without isotropic regularization. **Right:** Per-game improvement in AUC obtained by encouraging isotropic representation geometry. Across both algorithms, isotropic regularization improves final performance and sample efficiency.

Fig. 14 and Table 3 show that enforcing isotropic and Gaussian representations through the auxiliary SIGReg objective significantly narrows the gap between the baseline and Kronecker-factored optimizer. Importantly, this improvement is obtained without adding significant computational or memory overhead.

4.5. Full Atari Suite

We next evaluate isotropic Gaussian representations at scale on the full Atari benchmark using PQN. Fig. 6 reports per-game improvements in area under the learning curve (AUC) relative to a RAdam baseline. Encouraging isotropic Gaussian structure in the learned representations yields broad and consistent gains across the suite. Out of 57 games, 51 (89.5%) exhibit improved performance, with a mean AUC improvement of 889% and a median improvement of 138%. Crucially, these improvements are not driven by a small number of outlier environments, but are distributed across games with diverse dynamics, reward structures, and exploration challenges. These results demonstrate that isotropic Gaussian representations scale reliably to large and heterogeneous benchmarks, providing a simple and effective mechanism for improving both learning efficiency and final performance in deep RL.

4.6. Beyond Atari and PQN

Policy Gradient Methods (PPO). Motivated by the results that isotropic Gaussian representation leads to significant improvements in value-based methods (Section 4.5), we next evaluate whether similar geometric effects, and their benefits, extend to policy-gradient algorithms. We focus on Proximal Policy Optimization (PPO) (Schulman et al., 2017), which differs substantially from PQN in both optimization dynamics and update structure. Although PPO is often regarded as more stable, we observe that it still suffers from representation collapse and neuron dormancy under long training horizons and non-stationary data distributions (Moalla et al., 2024; Mayor et al., 2025). Across the full Atari suite, encouraging isotropic representation geometry leads to consistent improvements in IQM human-normalized performance and higher AUC in the majority of games (see Fig. 6). Importantly, stabilization strategies that operate at the optimization or architectural level, such as Kronecker-factored optimization or multi-skip residual connections, do not reliably transfer to PPO and can even degrade performance (see Fig. 14 and Table 3). By contrast, modestly encouraging isotropy at the representation level yields improvements without altering the PPO update rule or introducing algorithm-specific modifications. These results reinforce the interpretation that isotropic representations effectively address optimization issues under non-stationarity and that this effect generalizes beyond value-based methods.

Continuous Control in Isaac Gym. To assess whether the benefits of isotropic Gaussian representations extend beyond discrete control and pixel-based domains, we evaluate continuous control tasks in Isaac Gym (Makoviychuk et al., 2021b). These environments exhibit strong non-stationarity due to contact dynamics, evolving state dis-

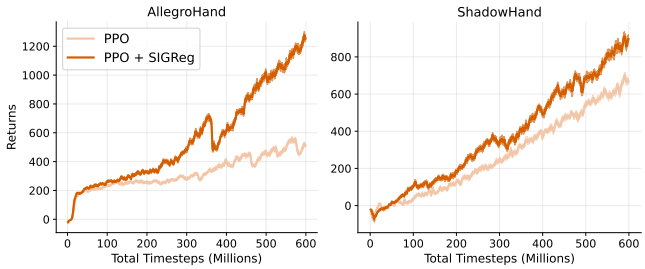

Figure 7: **Isaac Gym continuous control.** Learning curves on two representative locomotion tasks. Isotropic Gaussian representations improve stability and reduce variance. We report returns over 5 runs for each experiment. See Section D for additional results on four Isaac Gym control tasks.

tributions, and high-dimensional continuous action spaces. Across a range of locomotion and manipulation tasks, encouraging isotropic representation geometry leads to improved training stability and reduced variance across 5 random seeds as shown in Fig. 7.

5. Related Work

Deep RL agents operate under inherently non-stationary conditions, as both data distributions and learning targets evolve with the policy. This non-stationarity exacerbates representation drift and plasticity loss, leading to dormant neurons, rank collapse, and reduced effective capacity during training (Lyle et al., 2023; Obando-Ceron et al., 2025; Lyle et al., 2022; Nikishin et al., 2022; Moalla et al., 2024). Such degradation has been linked to performance collapse and diminished adaptability in long-horizon, continual, and online learning settings (Tang et al., 2025). Existing mitigation strategies typically rely on architectural changes, auxiliary losses, neuron reinitialization, or optimization-centric techniques (Nikishin et al., 2022; Liu et al., 2025c; Castanyer et al., 2025), as well as spectral and rank-based diagnostics (Lyle et al., 2023), but do not directly target the statistical structure of learned representations. A complementary line of work improves representation stability through auxiliary objectives inspired by self-supervised learning (Echchahed & Castro, 2025). These include contrastive methods such as CURL (Laskin et al., 2020), predictive approaches like SPR (Schwarzer et al., 2021), and metric-based objectives such as MiCO (Castro et al., 2021), which mitigate representation collapse and neuron dormancy (Kumar et al., 2021; Lyle et al., 2021; Sokar et al., 2023; Liu et al., 2025c; Obando-Ceron et al., 2025). See App. A for further discussion.

6. Conclusion

We studied deep RL through the lens of representation geometry and showed that isotropic Gaussian representations provide a principled and effective solution to instability under non-stationarity. Our analysis formalizes representation learning as a tracking problem with drifting targets and establishes that, under isotropic Gaussian embeddings, the zero tracking-error equilibrium is stable, with bounded and decreasing error dynamics. This theoretical result offers a clear explanation for why certain representation structures are intrinsically robust to evolving objectives. We evaluated SIGReg as a lightweight and practical mechanism for shaping representations toward this favorable geometry. Across controlled non-stationary supervised settings and large-scale deep RL benchmarks, SIGReg consistently improved training stability, reduced representation collapse and neuron dormancy, and led to substantial performance gains.

Discussion. A defining challenge of deep RL is that representation learning and control are inseparable: as policies improve, both data distributions and learning targets evolve, so representations are never trained against a fixed objective. Non-stationarity is therefore a structural property of deep RL, not a secondary source of noise. Most prior work addresses this challenge indirectly, for example through optimization stabilization or variance reduction, implicitly treating representations as passive byproducts of training. Our results suggest this view is incomplete: under non-stationary supervision, representation geometry plays a central role in determining learning stability. Viewing representation learning as a tracking problem with drifting targets clarifies why anisotropic or low-entropy embeddings are fragile. Such representations concentrate capacity along directions favored by transient targets, amplifying sensitivity to drift and accelerating the loss of effective capacity. From this perspective, representation collapse and neuron dormancy are predictable consequences of unconstrained geometry. This motivates representation-level inductive biases that remain well-conditioned as objectives evolve. Simple statistical constraints, such as isotropy and controlled variance, provide a robust foundation for stable and scalable learning under continual change.

Limitations. This work focuses on shaping the marginal distribution of learned representations and does not explicitly enforce task-specific structure or semantic alignment. While isotropic Gaussian representations provide a strong default prior under uncertainty and non-stationarity, they may be suboptimal for tasks requiring highly structured features, and balancing isotropy with task-adaptive biases remains an open challenge. Our analysis assumes simplified linear readouts and approximate stationarity of the representation covariance; extending these results to nonlinear heads, fully coupled off-policy actor–critic algorithms (Seo et al., 2025; Liu et al., 2025b), multi-task training (Willi et al., 2024; McLean et al., 2025), and continual deep RL settings (Tang et al., 2025) is an important direction for future work.

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A. Related Work

Non-Stationarity and Deep RL. Deep RL agents are trained under inherently non-stationary conditions, as both the data distribution and learning targets evolve with the policy. This non-stationarity exacerbates representation drift and plasticity loss, often resulting in dormant neurons and degraded learning dynamics (Lyle et al., 2023; Obando-Ceron et al., 2025). Recent empirical studies have shown that deep RL agents progressively lose effective capacity during training, manifested as a reduction in representation rank and an increasing fraction of inactive units (Lyle et al., 2022; Nikishin et al., 2022; Ceron et al., 2023; Moalla et al., 2024; Liu et al., 2025a). Such degradation has been linked to performance collapse and reduced adaptability in long-horizon, continual, and online learning settings (Tang et al., 2025).

Prior approaches to mitigate plasticity loss typically focus on architectural modifications, auxiliary losses, or explicit neuron recycling and reinitialization strategies (Ceron et al., 2024c; Sokar et al., 2025; Nikishin et al., 2022; Liu et al., 2025c). Other work has analyzed representational collapse and capacity loss through spectral and rank-based diagnostics, highlighting their prevalence across algorithms and domains (Lyle et al., 2023). Castanyer et al. (2025) showed that mitigating gradient degradation through approximate second-order optimization and multi-skip information propagation can alleviate plasticity loss, enabling more stable scaling of deep RL architectures. More recently, (Obando-Ceron et al., 2025) evaluated simplicial embeddings as a geometric inductive bias on latent representations, showing that structured and sparse feature spaces can stabilize critic bootstrapping and improve sample efficiency in actor–critic methods, alleviating feature collapse as a consequence of non-stationarity. While effective, these approaches primarily target optimization dynamics or architectural design, rather than directly shaping the statistical structure of learned representations.

Representation Learning in Deep RL. Learning stable and expressive representations is a long-standing challenge in deep RL. A substantial body of work has introduced auxiliary objectives to regularize representations and improve stability (Echchahed & Castro, 2025), often inspired by self-supervised learning (Balestriero et al., 2023). Contrastive methods such as CURL (Laskin et al., 2020) encourage feature diversity, while predictive approaches such as SPR (Schwarzer et al., 2021) promote temporal consistency through future-state prediction. Metric-based objectives (Desharnais et al., 2004; Ferns et al., 2012; Castro, 2020; Zang et al., 2023) such as MiCO (Castro et al., 2021) aim to align representational geometry with behavioral similarity.

More broadly, auxiliary losses and representation regularization techniques have been shown to mitigate representation collapse and neuron dormancy in deep RL (Kumar et al., 2021; Lyle et al., 2021; Sokar et al., 2023; Liu et al., 2025c; Obando-Ceron et al., 2025). While effective, these methods typically introduce additional prediction heads, task-specific objectives, or architectural complexity, and their benefits can be sensitive to hyperparameter choices and domain characteristics. For instance, Farebrother et al. (2023) propose *Proto-Value Networks*, which scale representation learning by training on a large family of auxiliary tasks derived from the successor measure. Similarly, Cetin et al. (2023) show that imposing an explicit *geometric prior* on the latent space, by learning representations in hyperbolic space, can improve performance and generalization. Fujimoto et al. (2025) pursue general-purpose model-free RL by leveraging learned representations inspired by model-based objectives that approximately linearize the value function, enabling competitive performance across diverse benchmarks with a single set of hyperparameters.

In contrast to auxiliary-task-based approaches (Gelada et al., 2019; Zhang et al., 2021; Fujimoto et al., 2023; Obando-Ceron et al., 2025), we focus on shaping representation geometry directly through a lightweight statistical regularizer. Our approach is inspired by recent advances in self-supervised learning, which demonstrate that enforcing simple statistical constraints, such as isotropy and Gaussianity, can be sufficient to yield stable representations under non-stationary targets. By encouraging isotropic Gaussian structure at the representation level, our method complements existing approaches while remaining simple, computationally efficient, and broadly applicable across deep RL algorithms.

B. Formal Analysis

B.1. In the presence of non-stationary tasks, Isotropic Gaussian makes zero tracking error a stable equilibrium

To simplify the analysis, we consider a linear critic on top of the embedding:

$$Q_\theta(s,a) := w^\top \phi(s,a) \tag{2}$$

where

- $\phi(s,a) \in \mathbb{R}^d$ is the **penultimate-layer embedding**

- $w \in \mathbb{R}^d$ is the last-layer weight vector

Define the TD target (the target is time-varying which is the source of non-stationarity):

$$y_t = r + \gamma Q_{\theta^-}(s', a') \tag{3}$$

The expected critic loss is

$$\mathcal{L}_t(w) = \mathbb{E}\left[\left(w^\top \phi - y_t\right)^2\right] \tag{4}$$

If we expand the loss term:

$$\mathcal{L}_t(w) = \mathbb{E}\left[\left(w^\top \phi - y_t\right)\left(\phi^\top w - y_t^\top\right)\right] \tag{5}$$

$$\mathcal{L}_t(w) = \mathbb{E}[w^\top \phi \phi^\top w] - 2\mathbb{E}[y_t \phi^\top w] + \mathbb{E}[y_t^2] \tag{6}$$

Define:

$$\Sigma_\phi(t) := \mathbb{E}[\phi \phi^\top] \qquad b_t := \mathbb{E}[\phi y_t] \tag{7}$$

where $\Sigma_\phi(t)$ is the covariance matrix of the embedding vectors and b_t is the drift term caused by non-stationarity. Then:

$$\boxed{\mathcal{L}_t(w) = w^\top \Sigma_\phi(t) w - 2w^\top b_t + \mathbb{E}[y_t^2]} \tag{8}$$

The gradient of the loss w.r.t the weight of the final layer:

$$\nabla_w \mathcal{L}_t(w) = 2\Sigma_\phi(t) w - 2b_t \tag{9}$$

the final term was eliminated due to assuming target weights to be independent of the actual weights. We analyze continuous-time gradient flow:

$$\boxed{\dot{w}(t) = -\nabla_w \mathcal{L}_t(w) = -2\Sigma_\phi(t) w + 2b_t} \tag{10}$$

At each time t, the instantaneous minimizer for the weight matrix of the final layer can be calculated as follows:

$$\nabla_w \mathcal{L}_t(w_t^*) = 0 \tag{11}$$

This gives:

$$\boxed{w_t^* = \Sigma_\phi(t)^{-1} b_t} \tag{12}$$

This optimal weight moves in time as a result of factors imposing non-stationarity. Our goal in this analysis is to:

1. Define the tracking error for the weights of last layer.
2. Derive the formula for how this tracking error changes.
3. Derive the energy, Lyapunov, function for the norm of tracking error and show that isotropic Gaussian structure makes the zero equilibrium stable, meaning that an increase in the norm of tracking error will be damped and converge to zero over time.

We define the tracking error at time t as the difference between the weight matrix and the optimal unknown weights:

$$\boxed{e(t) := w(t) - w_t^*} \tag{13}$$

Our goal is to analyze under what conditions $||e||_2^2 = 0$ is contractive. We focus on this equilibrium because it shows whether, when $||e||_2$ increases due to non-stationarity, it eventually returns to zero. To perform this analysis, we adopt Lyapunov stability analysis common in the analysis of the stability of equilibria of dynamical systems (Massera, 1949; Lefschetz & LaSalle, 1961). We choose a quadratic function as the Lyapunov function, which represents the energy of the system:

$$\boxed{\Gamma = ||e||_2^2}. \tag{14}$$

The condition for stability of $||e||_2^2 = 0$ is that the derivative of the Lyapunov function be negative, meaning that an increase in $||e||_2$, and therefore the energy of the system, leads to contraction of the dynamics to the equilibrium $||e||_2^2 = 0$.

Theorem B.1 (Tracking error dynamics)**.** *Assume that $\Sigma_\phi(t) \succ 0$ is constant over time (e.g., enforced by regularization) and that b_t is differentiable. Under gradient flow, the time derivative of Γ satisfies*

$$\dot{\Gamma} = -4\, e(t)^\top \Sigma_\phi(t)\, e(t) \;-\; 2\, e(t)^\top \Sigma_\phi(t)^{-1} \dot{b}_t. \tag{15}$$

Proof. To analyze the stability of $||e||_2^2 = 0$, we need the time derivative of tracking error term:

$$\dot{e} = \dot{w} - \dot{w}_t^* \tag{16}$$

Substitute $\dot{w}$:

$$\dot{e} = (-2\Sigma_\phi w + 2b_t) - \dot{w}_t^* \tag{17}$$

$$\dot{e} = -2\Sigma_\phi (e + w_t^*) + 2b_t - \dot{w}_t^* \tag{18}$$

Since $\Sigma_\phi w_t^* = b_t$:

$$\boxed{\dot{e} = -2\Sigma_\phi(t) e - \dot{w}_t^*} \tag{19}$$

We are interested in writing this dynamic formula in terms of the embedding and drift. To write $\dot{w}_t^*$ in terms of the embedding and drift, recall:

$$w_t^* = \Sigma_\phi^{-1} b_t \tag{20}$$

From matrix calculus (Petersen et al., 2008):

$$\frac{d}{dt}\Sigma^{-1} = -\Sigma^{-1}\dot{\Sigma}\Sigma^{-1}. \tag{21}$$

Thus:

$$\dot{w}_t^* = \Sigma_\phi^{-1}\dot{b}_t - \Sigma_\phi^{-1}\dot{\Sigma}_\phi\Sigma_\phi^{-1} b_t \tag{22}$$

Substitute into error dynamics:

$$\boxed{\dot{e} = -2\Sigma_\phi e - \Sigma_\phi^{-1}\dot{b}_t + \Sigma_\phi^{-1}\dot{\Sigma}_\phi w_t^*} \tag{23}$$

Differentiating the Lyapunov function:

$$\dot{\Gamma} = 2e^\top \dot{e}. \tag{24}$$

Substitute $\dot{e}$:

$$\dot{\Gamma} = -4e^\top\Sigma_\phi e - 2e^\top\Sigma_\phi^{-1}\dot{b}_t + 2e^\top\Sigma_\phi^{-1}\dot{\Sigma}_\phi w_t^*. \tag{25}$$

Writing all terms in terms of embedding, tracking error, and drift:

$$\boxed{\dot{\Gamma} = \underbrace{-4e^\top\Sigma_\phi e}_{\text{contraction}} \underbrace{-2e^\top\Sigma_\phi^{-1}\dot{b}_t}_{\text{target non-stationarity}} + \underbrace{2e^\top\Sigma_\phi^{-1}\dot{\Sigma}_\phi\Sigma_\phi^{-1}b_t}_{\text{representation drift}}} \tag{26}$$

Our goal is to show that by regularizing the embedding dimension, certain conditions lead to stability of $||e||_2^2 = 0$, or in other words, $\dot{\Gamma} < 0$. Since we assume regularizing the embedding distribution (leading to a fixed desired covariance matrix), we can remove the third term. □

B.1.1. First Term Analysis

Since the covariance matrix is positive definite, the sign of the first term is always negative. Therefore, this term acts as a contractor, adding a negative component to the Lyapunov function's derivative. This term will not cause a problem as it always increases the stability of the equilibrium.

B.1.2. SECOND TERM ANALYSIS

Although the sign in front of this term is negative, the whole term could be negative or positive. It depends on the angle between the tracking error and the change in the drift vector under the inverse of the covariance matrix as the metric for calculating the inner product. Therefore, it could be helpful (if negative) or harmful (if positive). Our analysis will focus mainly on limiting and bounding the norm of this term, since the sign is not under our control and we cannot exploit the cases in which this term is helpful.

B.1.3. THIRD TERM ANALYSIS

The third term is due to the drift in the embedding. As explained before, we are looking for the distribution that we need to choose as the desired distribution in the SIGReg regularizer. Therefore, we can remove this term since we regularize the embedding to be a fixed desired distribution, which makes $\dot{\Sigma}_\phi \approx 0$. Although the sign in front of this term is positive, this term could also be negative in multiple scenarios, for example if $\mathrm{Tr}(\Sigma_\phi)$ is being reduced, making $\dot{\Sigma}_\phi$ negative definite.

In the following subsections, we are going to consider only the first two terms, as we will regularize the distribution to be fixed and as a result covariance will not change, and assume that the $Tr(\Sigma_\phi) = c$. In other words, we have a fixed budget as the variance of the embedding.

B.1.4. WHY ISOTROPY IS HELPFUL?

To achieve a stable equilibrium, we want to ensure $\dot{\Gamma} < 0$ during training. This condition causes any non-zero tracking error to tend toward zero. Since the first two terms operate on different and independent vectors—the first term only on the tracking error vector and the second term on both the tracking error and drift—our only solution is to control the magnitude of these terms separately.

The first term (contraction) is always negative and helpful for driving the derivative toward negative values, since the covariance matrix is a positive semi-definite matrix. However, we want this term to be large and negative for all possible tracking error vectors.

Assume $\Sigma_\phi \succ 0$ and fix the total variance

$$\mathrm{Tr}(\Sigma_\phi) = \sum_{i=1}^{d} \lambda_i = c \tag{27}$$

where λ_i are the eigenvalues of Σ_ϕ. We always have the bound

$$e^\top \Sigma_\phi e \;\geq\; \lambda_{\min}(\Sigma_\phi)\, \|e\|_2^2 \tag{28}$$

Equality happens when the tracking error is aligned with the eigenvector corresponding to the smallest eigenvalue. We want to control this and ensure there is no direction for the tracking error in which the lower bound is small, as for that direction the contraction term becomes weak and small. Since the average of numbers is always greater than or equal to the smallest number:

$$\lambda_{\min}(\Sigma_\phi) \;\leq\; \frac{1}{d}\sum_{i=1}^{d} \lambda_i = \frac{c}{d} \tag{29}$$

it follows that

$$\min_{\|e\|_2=1} e^\top \Sigma_\phi e = \lambda_{\min}(\Sigma_\phi) \;\leq\; \frac{c}{d} \tag{30}$$

Therefore,

$$\max_{\Sigma_\phi:\, \mathrm{tr}(\Sigma_\phi)=c} \; \min_{\|e\|_2=1} e^\top \Sigma_\phi e = \frac{c}{d} \tag{31}$$

and equality holds if and only if:

$$\lambda_1 = \lambda_2 = \cdots = \lambda_d = \frac{c}{d} \tag{32}$$

In simple terms, we want to boost the direction with the smallest eigenvalue to ensure no direction will significantly dampen the contraction term. Since the average of eigenvalues is fixed and the smallest eigenvalue will always be less than or equal

to the average, the best case is when the smallest eigenvalue equals the average, which occurs when all eigenvalues are the same.

For the second term, since two different vectors are involved, we cannot use the positive semi-definiteness of the covariance matrix. Therefore, this term could be positive or negative and could cause the whole derivative to be large and positive. To control the worst-case scenario, we consider the upper bound for $|e^\top \Sigma_\phi^{-1} \dot{b}_t|$. Since $b_t = \Sigma_t w_t^*$, assuming negligible shift in the covariance matrix:

$$|e^\top \Sigma_\phi^{-1} \dot{b}_t| \leq \lambda_{\max}(\Sigma_\phi^{-1}) \lambda_{\max}(\Sigma_\phi) \|e\|_2 = \frac{\lambda_{\max}(\Sigma_\phi)}{\lambda_{\min}(\Sigma_\phi)} \|e\|_2 = \kappa(\Sigma_\phi) \|e\|_2 \tag{33}$$

where $\kappa(\Sigma_\phi)$ is the condition number of the covariance matrix. This term clearly shows that to minimize this upper bound, we need to set the condition number to the minimum value possible, which is one. This will be achieved if and only if the distribution is isotropic: $\lambda_{\max} = \lambda_2 = \cdots = \lambda_{\min} = \frac{c}{d}$.

B.1.5. Among Isotropic Distributions, Gaussian Leads to Minimum Fluctuations in the Drift Term

To justify the choice of Gaussian as the distribution instead of other isotropic distributions, we focus on the second term in which the time derivative of $b_t = \mathbb{E}[\phi y_t]$ is involved. We show that if the distribution is non-Gaussian, some extra terms appear that increase the variance of the second term. As a result, the uncertainty of the sign of the second term increases, which is not desirable. Before showing this, we need to establish some properties of general and Gaussian random variables.

If embedding vectors ϕ follow Gaussian distribution (proof of stein's lemma). Let $\phi \sim \mathcal{N}(0, \Sigma_\phi)$ with density

$$p(\phi) = \frac{1}{Z} \exp\Big(-\tfrac{1}{2} \phi^\top \Sigma_\phi^{-1} \phi\Big) \tag{34}$$

Then

$$\nabla_\phi p(\phi) = -\Sigma_\phi^{-1} \phi \, p(\phi) \tag{35}$$

$$\boxed{\phi \, p(\phi) = -\Sigma_\phi \nabla_\phi p(\phi)} \tag{36}$$

For any smooth function $f : \mathbb{R}^d \to \mathbb{R}$,

$$\mathbb{E}[\phi f(\phi)] = \int_{\mathbb{R}^d} \phi f(\phi) \, p(\phi) \, d\phi \tag{37}$$

Substitute Eq. 36 in Eq. 37:

$$\mathbb{E}[\phi f(\phi)] = -\Sigma_\phi \int_{\mathbb{R}^d} f(\phi) \, \nabla_\phi p(\phi) \, d\phi \tag{38}$$

If we write the integral dimension-wise and apply integration by part:

$$\int f(\phi) \, \nabla_\phi p(\phi) \, d\phi = \sum_{j=1}^{d} \int f(\phi) \, \frac{\partial p(\phi)}{\partial \phi_j} \, d\phi \tag{39}$$

$$\int f(\phi) \, \frac{\partial p(\phi)}{\partial \phi_j} \, d\phi = \Big[f(\phi) \, p(\phi) \Big]_{\phi_j = -\infty}^{\phi_j = \infty} - \int p(\phi) \, \frac{\partial f(\phi)}{\partial \phi_j} \, d\phi \tag{40}$$

As we have assumed multivariate Gaussian distribution, each dimension will have marginal uni variate distribution which decays exponentially fast. If we assume that $f(\phi)$ increases at most polynomially fast, we can conclude that:

$$\lim_{\|\phi\|_2 \to \infty} f(\phi) p(\phi) = 0 \tag{41}$$

so the boundary term vanishes:

$$\Big[f(\phi) \, p(\phi) \Big]_{\phi_j = -\infty}^{\phi_j = \infty} \approx 0 \tag{42}$$

Substituting back into Eq. 38:

$$\mathbb{E}[\phi f(\phi)] = \Sigma_\phi \int_{\mathbb{R}^d} p(\phi)\, \nabla_\phi f(\phi)\, d\phi \tag{43}$$

Therefore,

$$\boxed{\mathbb{E}[\phi f(\phi)] = \Sigma_\phi\, \mathbb{E}[\nabla_\phi f(\phi)]} \tag{44}$$

General form for an arbitrary distribution. Let $p(\phi)$ be any smooth density and define residual r as:

$$r(\phi) := \phi + \Sigma_\phi \nabla_\phi \log p(\phi) \tag{45}$$

This residual function is equal zero if and only if the distribution is Gaussian as in that case:

$$r(\phi) := \phi + \Sigma_\phi \Sigma_\phi^{-1} \phi = 0 \tag{46}$$

If we multiply both sides by $f(\phi)$ and take expectation:

$$\mathbb{E}[r(\phi) f(\phi)] = \mathbb{E}[\phi f(\phi)] + \Sigma_\phi \mathbb{E}[f(\phi) \nabla_\phi \log p(\phi)] \tag{47}$$

To simplify the second term, recall that:

$$\nabla_\phi \log p(\phi) = \frac{\nabla_\phi p(\phi)}{p(\phi)}, \tag{48}$$

so that

$$\mathbb{E}[f(\phi)\, \nabla_\phi \log p(\phi)] = \int f(\phi)\, \nabla_\phi p(\phi)\, d\phi. \tag{49}$$

By combining Eq. 38, 47, and 49 and reordering terms:

$$\boxed{\mathbb{E}[\phi f(\phi)] = \Sigma_\phi\, \mathbb{E}[\nabla_\phi f(\phi)] + \mathbb{E}[f(\phi)\, r(\phi)]} \tag{50}$$

If the distribution is Gaussian, the second term in Eq. 50 becomes zero and we recover the stein's lemma for multivariate Gaussian distribution.

Effect on the second term in the derivative of Lyapanouv formula. Recall

$$\boxed{\dot{\Gamma} = \underbrace{-4e^\top \Sigma_\phi e}_{\text{contraction}} \underbrace{-2e^\top \Sigma_\phi^{-1} \dot{b}_t}_{\text{target non-stationarity}} \underbrace{+2e^\top \Sigma_\phi^{-1} \dot{\Sigma}_\phi \Sigma_\phi^{-1} b_t}_{\text{representation drift}}} \tag{51}$$

and

$$b_t = \mathbb{E}[\phi\, y_t(\phi)]. \tag{52}$$

Our goal is to ensure $\dot{\Gamma}$ is always, or most of the time, negative. Since we are regularizing the embedding to have a fixed distribution, the third term will be negligible. The first term is also always negative and therefore always helpful. Here we show that a non-Gaussian distribution causes the variance of the second term to increase. As a result, the sign of this term may fluctuate frequently, causing this term to be positive and problematic.

For a general embedding distribution,

$$b_t = \Sigma_\phi\, \mathbb{E}[\nabla_\phi y_t(\phi)] + \mathbb{E}[y_t(\phi)\, r(\phi)] \tag{53}$$

Differentiating with respect to time:

$$\dot{b}_t = \Sigma_\phi \frac{d}{dt} \mathbb{E}[\nabla_\phi y_t(\phi)] + \frac{d}{dt} \mathbb{E}[y_t(\phi)\, r(\phi)] \tag{54}$$

Substitute into the target non-stationarity Lyapunov term:

$$-2e^\top \Sigma_\phi^{-1} \dot{b}_t = -2e^\top \frac{d}{dt} \mathbb{E}[\nabla_\phi y_t(\phi)] - 2e^\top \Sigma_\phi^{-1} \frac{d}{dt} \mathbb{E}[y_t(\phi)\, r(\phi)] \tag{55}$$

Since we have considered the Q-values to be calculated using a linear head, the first term in Eq. 55 is independent of the choice of distribution. Therefore, if the distribution is Gaussian, the only term present will be the first one, and the variance of the whole term will be reduced. In the case of a non-Gaussian distribution, the second term has a variance that makes the sign of the second term in the Lyapunov equation fluctuate and cause instability.

The variance of the residual term due to usage of a non-Gaussian distribution is proportional to:

$$\mathrm{Var}\left(e^\top \Sigma_\phi^{-1} \frac{d}{dt}\mathbb{E}[y_t(\phi)\, r(\phi)]\right) \;\propto\; \mathbb{E}\left[\left(e^\top \Sigma_\phi^{-1} y_t(\phi)\, r(\phi)\right)^2\right] \tag{56}$$

Recall that:

$$r(\phi) := \phi + \Sigma_\phi \nabla_\phi \log p(\phi) \tag{57}$$

Residual term is a non-linear function of ϕ due to $\nabla_\phi \log p(\phi)$. If we write the Taylor expansion of residual vector around $\bar{r} = \mathbb{E}[r]$:

$$\mathrm{Var}\left(e^\top \Sigma_\phi^{-1} \frac{d}{dt}\mathbb{E}[y_t(\phi)\, r(\phi)]\right) \;\propto\; \mathbb{E}\Big[\big(e^\top \Sigma_\phi^{-1} y_t(\phi)\big)^2 \big(J_r(\bar{r})(\phi-\bar{r}) + \frac{1}{2}H_r(\bar{r})[\phi-\bar{r},\phi-\bar{r}] + \cdots\big)^2\Big] \tag{58}$$

From this formula, high-order moments of ϕ emerge, which result in a positive value for the variance of the residual term, which in turn increases the total variance of the second term of the derivative of the Lyapunov function. Therefore, a non-Gaussian distribution will lead to a high-variance drift term that may lead to instability. The minimum possible variance can be achieved by a Gaussian distribution.

C. Ablation Studies and Additional Results

C.1. Non-Stationary CIFAR-10 Experiments with Different Optimizers

In this section, we extend the non-stationary CIFAR-10 experiments presented in subsection 4.1. We analyze the effect of different optimizers on training accuracy, feature rank, and neuron dormancy, and investigate how these quantities change after minimizing the SIGReg loss. The optimizers considered are:

- **Adam** (Kingma & Ba, 2017): a widely used first-order optimizer.
- **RAdam** (Liu et al., 2021): the default optimizer for PQN and used in some PPO implementations (Castanyer et al., 2025).
- **Kronecker-factored Optimizer (Kron)**(Martens & Grosse, 2015): a second-order optimizer shown to improve stability across various deep RL algorithms (Castanyer et al., 2025).

Method	Train Accuracy(↑)	SIGReg Loss(↓)	Feature Rank (↑)	Dormant Neurons[%](↓)
Adam	33.0	49.1	149.0	18.5
Adam+SIGReg	50.0	7.6	262.9	0.4
RAdam	29.8	42.3	148.8	12.6
RAdam+SIGReg	42.8	8.4	251.8	0.6
Kron	62.6	28.5	167.4	7.9
Kron+SIGReg	72.7	7.2	252.2	0.9

Table 1: **Non-stationary CIFAR-10.** Area Under the Curve (AUC) for different methods across evaluation metrics. SIGReg loss minimization improve results across all metrics and optimizers.

Figure 8 illustrates the effect of label shuffling on various metrics. When labels are shuffled, we observe: (1) significant drop in accuracy with poor recovery, (2) sharp increases in the SIGReg loss indicating loss of isotropic Gaussian structure, (3) collapse in feature rank, and (4) increased neuron dormancy. Notably, Kronecker-factored optimization implicitly reduces the SIGReg loss compared to first-order methods. However, explicitly minimizing the SIGReg objective further improves all optimizers by accelerating recovery, maintaining feature rank, and reducing dormancy.

Table 1 quantifies these observations through Area Under Curve measurements. Across all optimizers, adding SIGReg regularization leads to substantial improvements: training accuracy increases by 17–51%, SIGReg loss decreases by 78–85%, feature rank improves by 51–76%, and neuron dormancy drops by 89–98%. These results confirm that maintaining isotropic Gaussian structure is crucial for plasticity and recovery under non-stationary conditions.

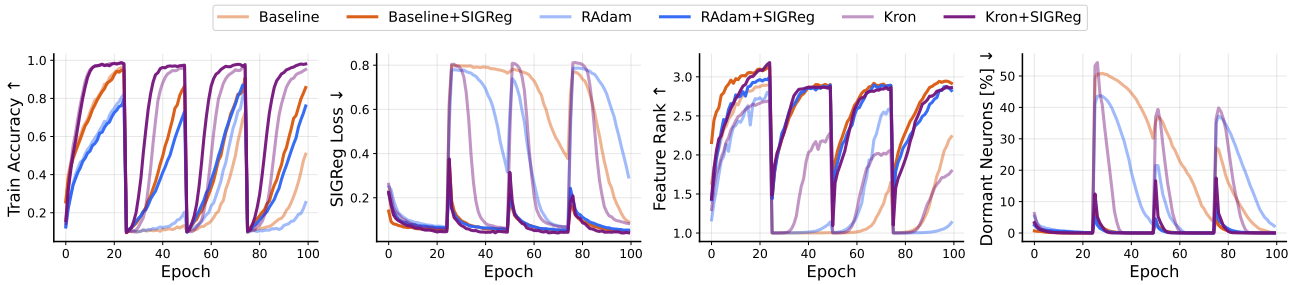

Figure 8: **Effect of non-stationarity and SIGReg loss minimization on representation stability and accelerating recovery in non-stationary CIFAR-10 experiment.** Non-stationarity induced by label shuffling causes a sharp drop in accuracy, increase in SIGReg loss, collapse of feature rank, and higher neuron dormancy. It can be observed that Kronecker-factored optimizer implicitly lowers the SIGReg loss compared to first-order methods. Explicitly minimizing SIGReg further improves all optimizers by accelerating recovery, preserving feature rank, and reducing neuron dormancy.

C.2. Effect of Different Target Distributions

In this section, we expand the results discussed in subsection 4.3. Results of this section are based on Atari-10 benchmark (Aitchison et al., 2023). Table 2 summarizes the impact of different representation regularization strategies on performance across Atari-10 games. We report both the fraction of games in which a method improves over the baseline and the average AUC improvement.

C.2.1. Alternative Isotropic Distributions

The goal of this subsection is to empirically evaluate whether Gaussianity of the target embedding distribution is an important factor for performance. To test this, we compare the Gaussian target to two alternative isotropic distributions with heavier tails, shown in the top section of Table 2. These distributions were chosen because, according to our formal analysis, heavier tails can increase instability. Across both PQN and PPO, the Gaussian target consistently outperforms the alternatives, both in terms of the number of games showing improvement and the average AUC increase across all 10 games. This indicates that matching the Gaussian distribution's characteristics, including tail behavior, contributes positively to performance.

To further isolate the effect of Gaussianity, we also evaluate a simple covariance whitening method, reported in the bottom section of Table 2. Whitening enforces isotropy by setting the covariance matrix to the identity, but it does not control tail characteristics. While this method achieves some improvements, particularly in PPO, it underperforms compared to the full Gaussian target in PQN and is generally less consistent. These results suggest that isotropy alone provides partial benefits, while Gaussianity, including its tail properties, plays a key role in stabilizing representations and achieving optimal performance.

C.2.2. Role of Symmetry and Tail Decay

In this subsection, we analyze the effect of symmetry and tail characteristics of the representation separately. The SIGReg loss is based on projecting high-dimensional embedding points onto one-dimensional directions and matching the estimated characteristic function of the projections to a target distribution. Since the characteristic function is the Fourier transform of the probability distribution, the target or projected embeddings can be complex numbers. The imaginary part captures all odd-order moments of the distribution, so if it is zero, the distribution is symmetric. The real part correlates with the even-order moments, which correspond to the tail behavior.

For a Gaussian target, which is symmetric, the characteristic function is real. However, the estimated characteristic function of the embeddings may have a nonzero imaginary component. The middle section of Table 2 shows the effect of minimizing each part separately. The results indicate that minimizing the real component is significantly more important than minimizing the imaginary component. In PQN, focusing on the real part even increases the number of games improved compared to minimizing both components simultaneously. The underlying reason is not fully understood, but it may result from an implicit reduction of the imaginary component when minimizing the real part.

Method	PQN		PPO	
	% of Games Improved (↑)	Avg. Improvement (↑)	% of Games (↑)	Avg. Improvement (↑)
Gaussian (Real and Imaginary parts)	90%	305%	70%	7.9%
Laplacian	90%	234%	60%	3.2%
Logistic	90%	209%	50%	-0.8%
Real Part Only	100%	151%	60%	2.4%
Imaginary Part Only	70%	56%	50%	1.9%
Covariance Whitening	90%	144%	70%	4.4%

Table 2: **Ablation study on Atari-10.** Percentage of games improved over baseline and average AUC improvement across all games. **Top:** different isotropic target distributions (Laplacian and Logistic). **Middle:** minimizing different SIGReg loss components. **Bottom:** whitening the covariance matrix without controlling tails. Gaussian with real and imaginary components performs best, while enforcing isotropy alone can improve some environments. In PQN, minimizing only the real part improves more games than the baseline Isotropic Gaussian, though with significantly smaller average improvement.

C.3. More Results on Implicit Isotropy in Stabilization Methods

In this section, we expand on the discussion in subsection 4.4 by investigating the connection between isotropic Gaussian representations and the stabilization mechanisms introduced by Castanyer et al. (2025), namely Kronecker-factored optimization and multi-skip residual architectures, both designed to improve stability in deep RL at scale.

Fig. 9 presents results for Parallelized Q-Networks (PQN) across the Atari-10 benchmark. Across nearly all games, Kronecker-factored optimization produces representations that are substantially closer to an isotropic Gaussian distribution, as indicated by lower SIGReg loss, compared to the baseline. Importantly, this effect emerges **without any explicit representation regularization**, suggesting that Kronecker-factored optimization implicitly encourages embeddings to adopt a geometry that is both Gaussian-like and approximately isotropic. Similarly, multi-skip residual architectures, originally designed to stabilize gradient propagation, also implicitly promote isotropic Gaussian embeddings, as shown in Fig. 10. These effects are not limited to PQN: analogous trends are observed in Proximal Policy Optimization (PPO), as illustrated in Fig. 11 and Fig. 12. Across both algorithms, the figures reveal a strong correlation between an increase in feature rank, a decrease in the percentage of dormant neurons, and an implicit reduction in SIGReg loss, highlighting that these stabilization mechanisms partially act by shaping the geometry of the learned representations. Another interesting observation from Figures 9 to 12 is that, across all games, PPO exhibits a lower SIGReg loss compared to PQN. Understanding the underlying reason for this difference remains an interesting research question.

Although Kronecker-factored optimization is effective and provides substantial improvements, it introduces additional memory and computational overhead due to the estimation of gradient curvature. This raises the question of whether similar benefits could be achieved using first-order optimizers if the geometry of the representations is explicitly controlled. Based on the patterns observed in Fig. 9, we hypothesize that a significant portion of the stability and performance gains from these methods arises from the implicit shaping of representation geometry. To test this hypothesis, we explicitly enforce isotropic Gaussian embeddings in baseline models using the auxiliary SIGReg objective. Fig. 14 and Table 3 show that this approach significantly narrows the performance gap relative to models trained with Kronecker-factored optimization, across both PQN and PPO. Notably, this improvement is achieved without introducing additional memory or computational burden compared to the baseline, highlighting the importance of embedding geometry rather than optimizer complexity.

Fig. 14 presents IQM human-normalized results for different methods. The upper panel shows PQN results, where explicit SIGReg loss minimization reduces the gap between the second-order Kronecker-factored optimizer and the first-order RAdam baseline. Adding multi-skip residual connections achieves nearly the same performance while being significantly faster than Kronecker-factored optimization (see the steps per second (SPS) column in Table 3). The bottom panel shows PPO results, indicating even more promising trends: SIGReg loss minimization on top of a first-order optimizer, with or without multi-skip connections, outperforms the second-order optimizer while remaining faster. Examining Table 3, we see that explicit SIGReg loss minimization successfully reduces the gap between Kronecker-factored optimization and the baseline, and can even surpass it, while being around ten times faster in PQN and three times faster in PPO, without requiring a memory bank for gradient curvature estimation.

Together, these observations suggest that representation isotropy and Gaussianity are key factors underlying the empirical stability gains seen with advanced stabilization mechanisms. Explicitly regularizing embeddings to be isotropic and Gaussian

	PQN			PPO		
Method	**% of Games Improved (↑)**	**Avg. Improvement (↑)**	**Avg. SPS (↑)**	**% of Games Improved (↑)**	**Avg. Improvement (↑)**	**Avg. SPS (↑)**
Kron	98.2%	1404%	1.6k	45.6%	-2.6%	1.8k
Baseline + SIGReg	89.5%	889%	9.3k	68.4%	25.5%	2.7k
Baseline + MultiSkip + SIGReg	92.9%	1695%	11.4k	87.7%	88.6%	3.2k

Table 3: **PQN and PPO Performance Across Full Atari Suite.** We report the percentage of games in which each method improves over the baseline, the average percentage of AUC improvement across all games, and the average steps per second (Avg. SPS). In terms of average AUC improvement, the baseline augmented with multi-skip residual connections and SIGReg loss minimization outperforms all other methods, while also achieving higher SPS and avoiding the memory overhead associated with computing gradient curvature. In terms of the percentage of games improved, Kronecker-factored optimization achieves the best result by a small margin in PQN, and it is outperformed by both two other cases in PPO. Overall, these results indicate that the performance gains provided by Kron do not justify the additional memory and computational cost.

can capture much of the benefit of second-order optimizers while retaining the efficiency of first-order methods.

C.4. Comprehensive Atari Benchmark Results

We evaluate our method on the Atari benchmark across 57 games using both PQN and PPO algorithms. This comprehensive evaluation allows us to assess the generalization of our approach across diverse environments and training paradigms. The second row of Table 3 summarizes the performance of SIGReg loss minimization across the full Atari suite. For PQN, applying SIGReg loss minimization improves performance in approximately 90% of games, with an average AUC improvement of 889%. For PPO, 68% of games show improvement, with an average AUC gain of 25%. This large difference between PQN and PPO can be attributed to two factors: (1) PPO exhibits a lower SIGReg loss compared to PQN in the baseline (see baseline results in Fig. 9 and Fig. 11), and (2) PPO already achieves strong performance, leaving limited room for further improvement. For per-game improvements and IQM human-normalized results, see Fig. 6.

In the following sections, we first demonstrate the effect of SIGReg loss minimization on several evaluation metrics that are useful for assessing the stability of deep RL methods. We then present per-game episode reward plots to illustrate the learning dynamics across all games.

C.4.1. Impact on Representation Quality

Figure 13 extends Fig. 4 to all games in the Atari-10 benchmark and considers two regularization factors, 10 and 0.2. It shows how reward, SIGReg loss, feature rank, and the percentage of dormant neurons change over time, with and without SIGReg regularization. The orange line corresponds to the baseline without SIGReg, while the other lines show different regularization strengths, where a larger λ means stronger regularization.

Across all games and settings, SIGReg loss minimization consistently reduces neuron dormancy, maintains a higher feature rank, and improves reward. The increase in rank is expected, since enforcing isotropy spreads variance more evenly across all principal components. However, the strong reduction in the percentage of dormant neurons is less straightforward and remains an interesting direction for future research.

C.4.2. Episode Reward for All Individual Games in Full Atari Suite

Figures 15 and 16 show the episode reward over time for individual games in PQN and PPO, respectively. Each plot includes three lines corresponding to different regularization factors. The line with $\lambda = 0$ represents the baseline method without SIGReg loss minimization. Improvements from SIGReg loss minimization are more pronounced in PQN. One possible reason is that, in PPO, the baseline SIGReg loss is already lower than in PQN, leaving less room for improvement. Additionally, the baseline performance in PPO is significantly higher than in PQN, further limiting the potential for noticeable gains.

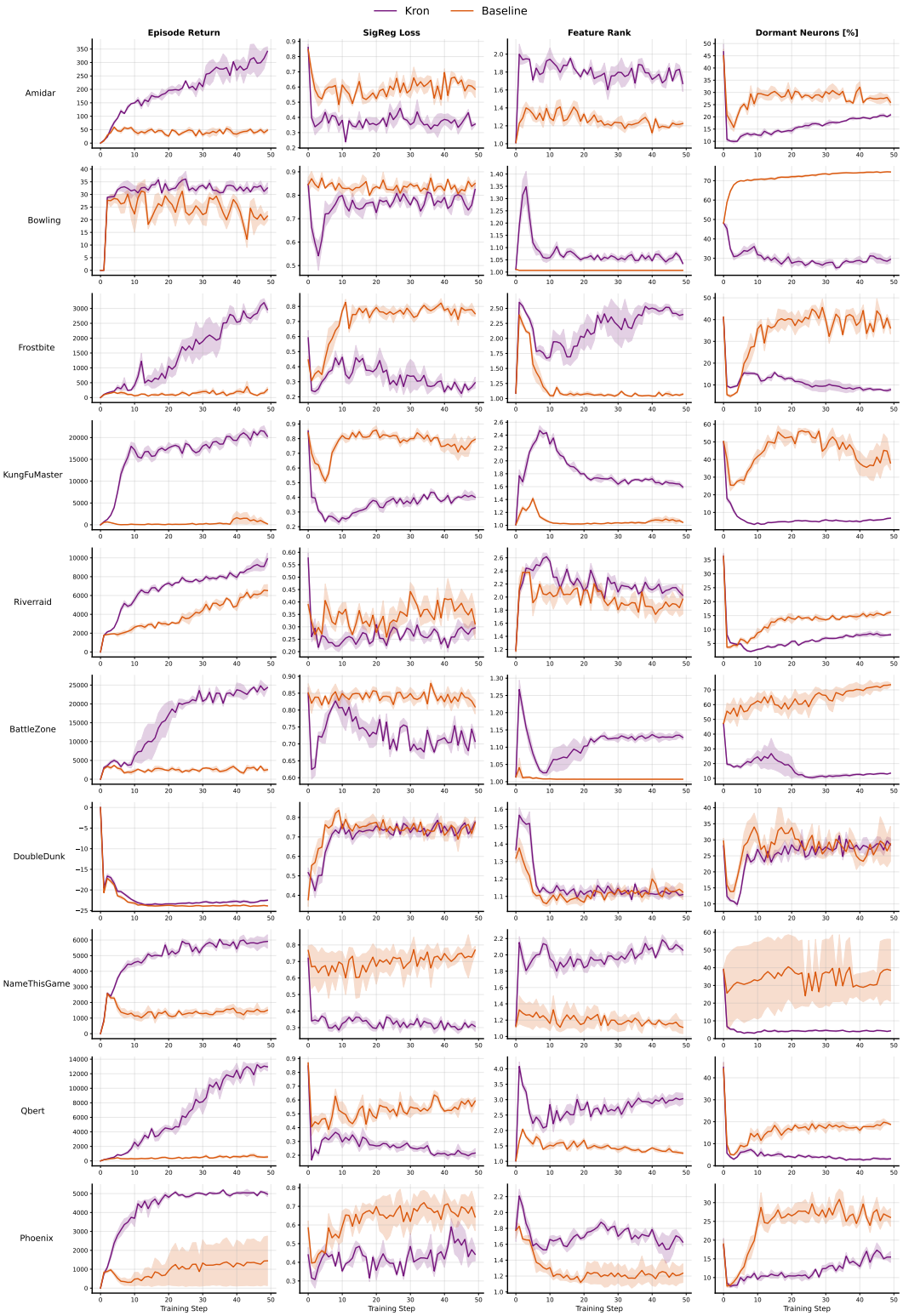

Figure 9: **Atari-10, PQN, Kron Optimizer.** The change of reward, SIGReg loss, feature rank, and percentage of dormant neurons. In almost all games, Kron leads to implicit minimization of SIGReg loss, higher rank, and lower dormancy.

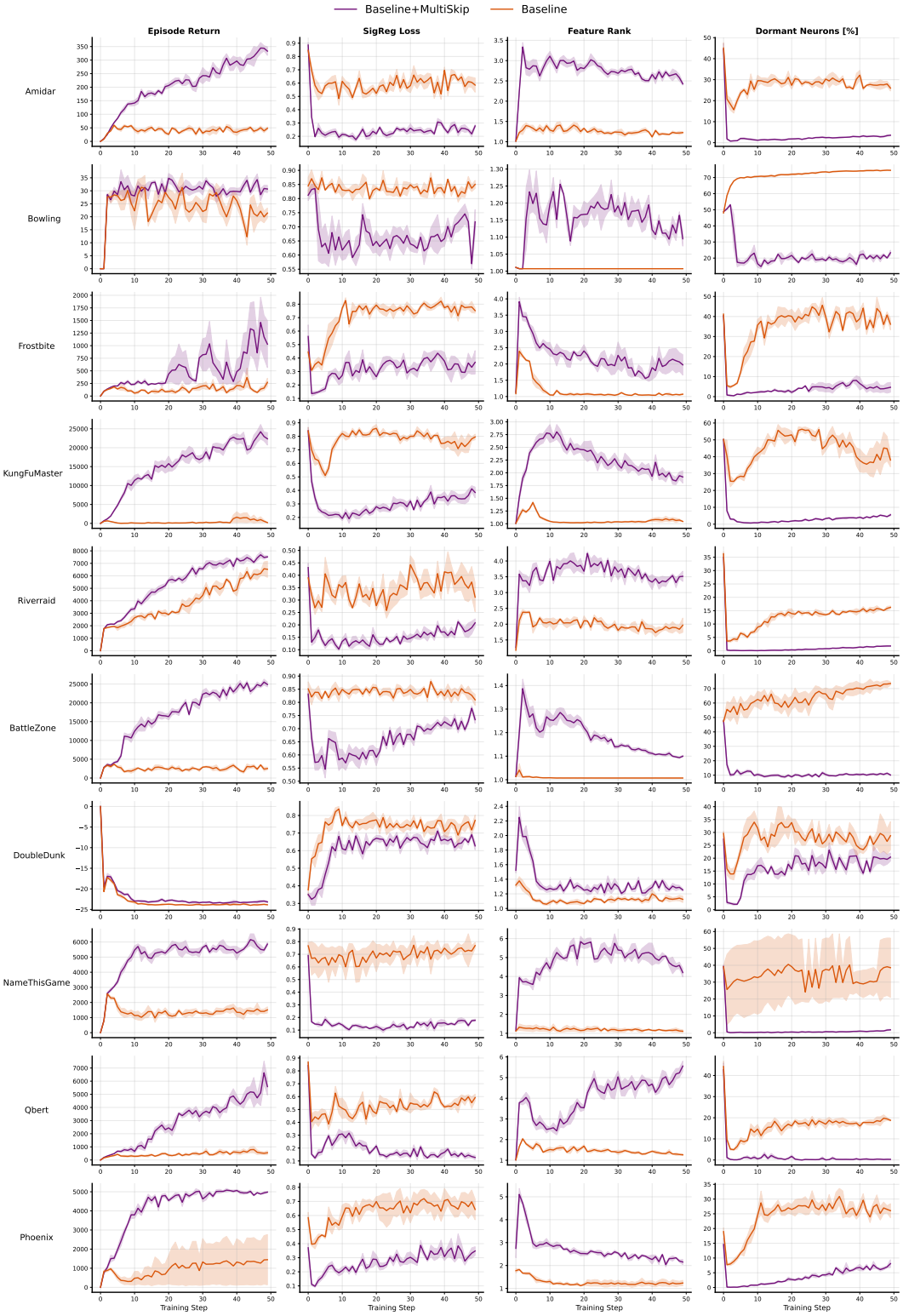

Figure 10: **Atari-10, PQN, Multi-skip Residual Architecture.** The change of reward, SIGReg loss, feature rank, and percentage of dormant neurons. In almost all games, using Multi-skip Residual Architecture leads to implicit minimization of SIGReg loss, higher rank, and lower dormancy.

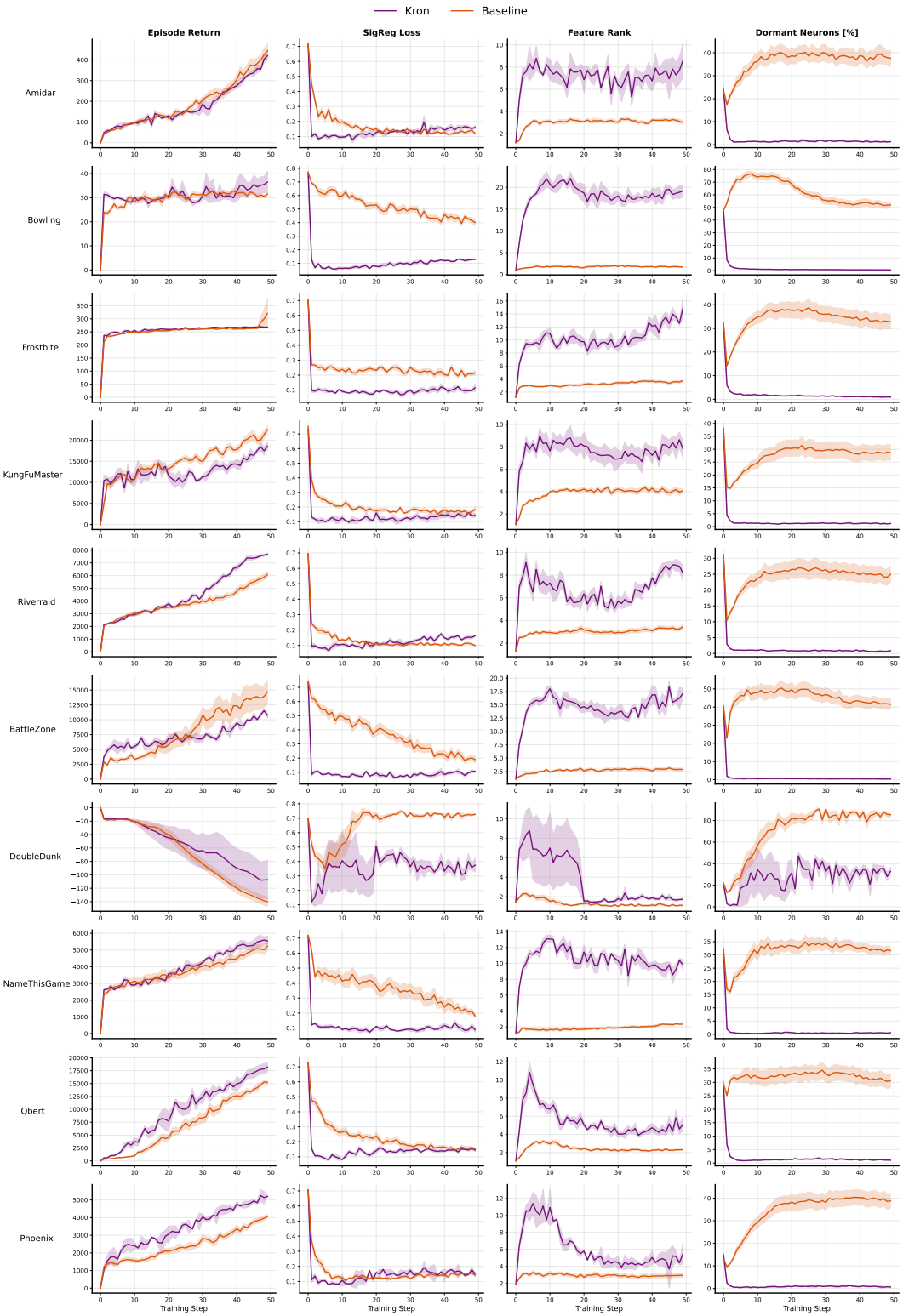

Figure 11: **Atari-10, PPO, Kron Optimizer.** The change of reward, SIGReg loss, feature rank, and percentage of dormant neurons. In almost all games, Kron leads to implicit minimization of SIGReg loss, higher rank, and lower dormancy.

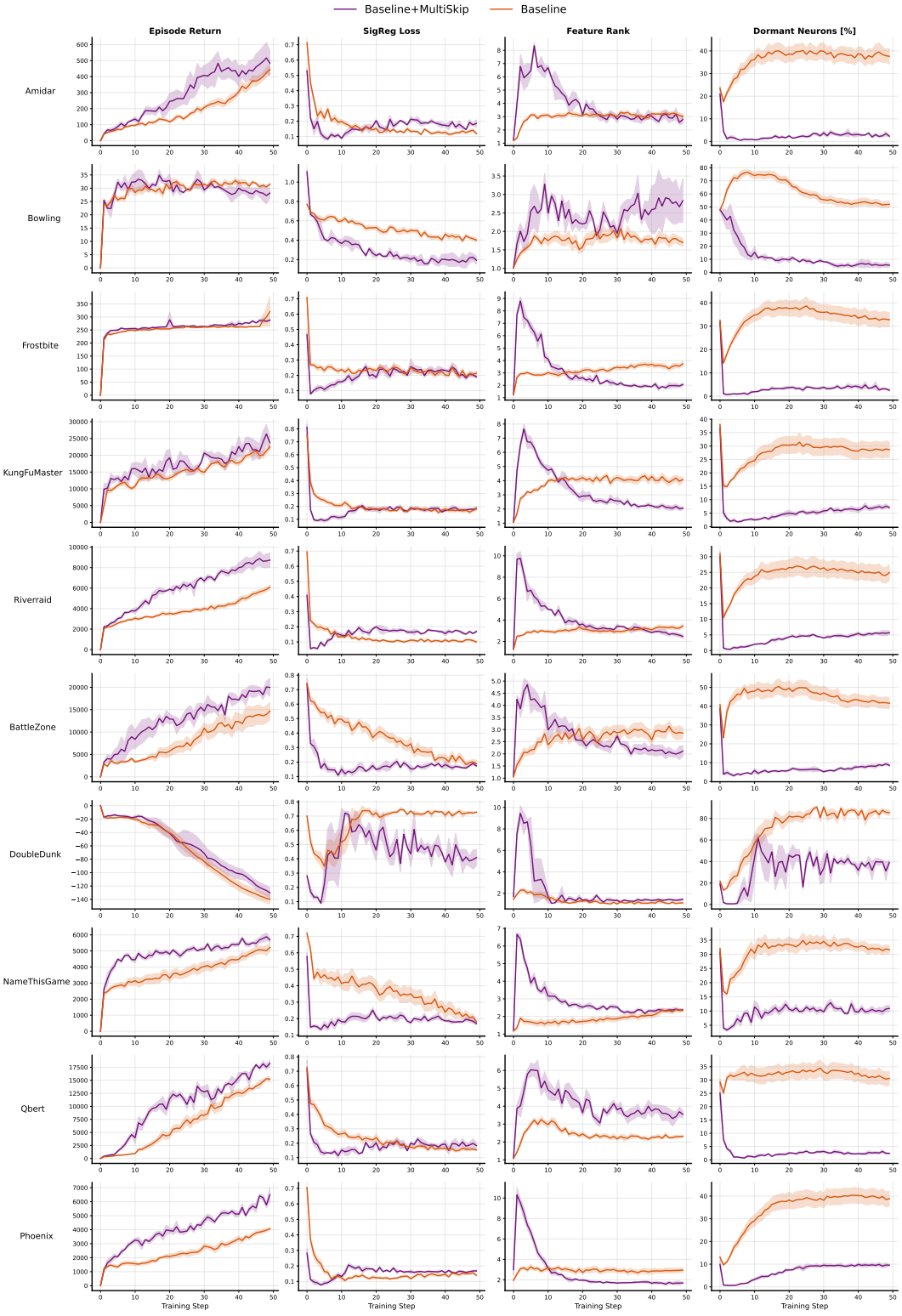

Figure 12: **Atari-10, PPO, Multi-skip Residual Architecture.** The change of reward, SIGReg loss, feature rank, and percentage of dormant neurons. In almost all games, using Multi-skip Residual Architecture leads to implicit minimization of SIGReg loss, higher rank, and lower dormancy.

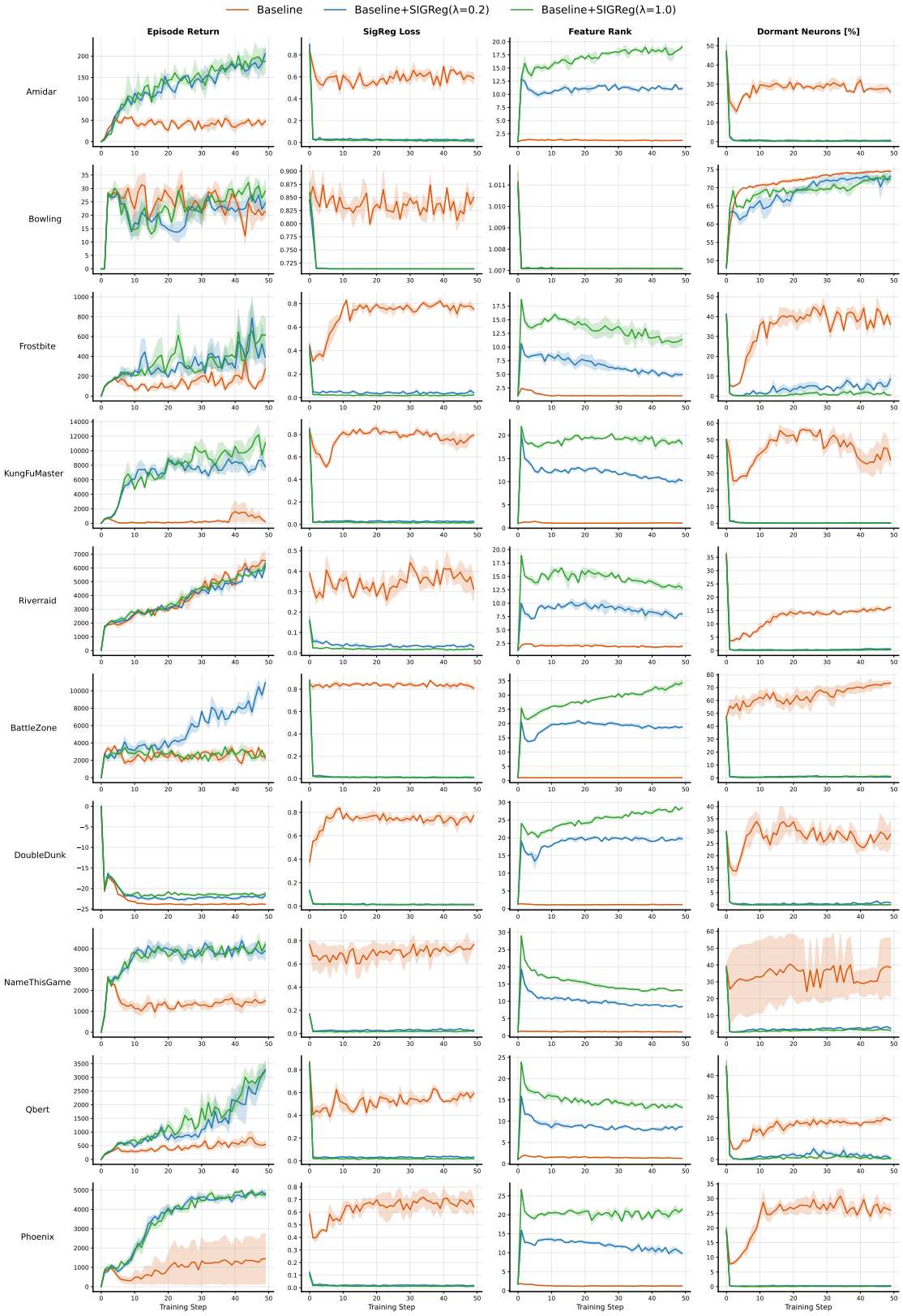

Figure 13: **Atari-10, PQN with Explicit SIGReg Loss Minimization.** The change of reward, SIGReg loss, feature rank, and percentage of dormant neurons through time. The orange line is the baseline without SIGReg loss minimization, and the other lines are SIGReg loss minimization with different strengths (Larger λ means stronger regularization).

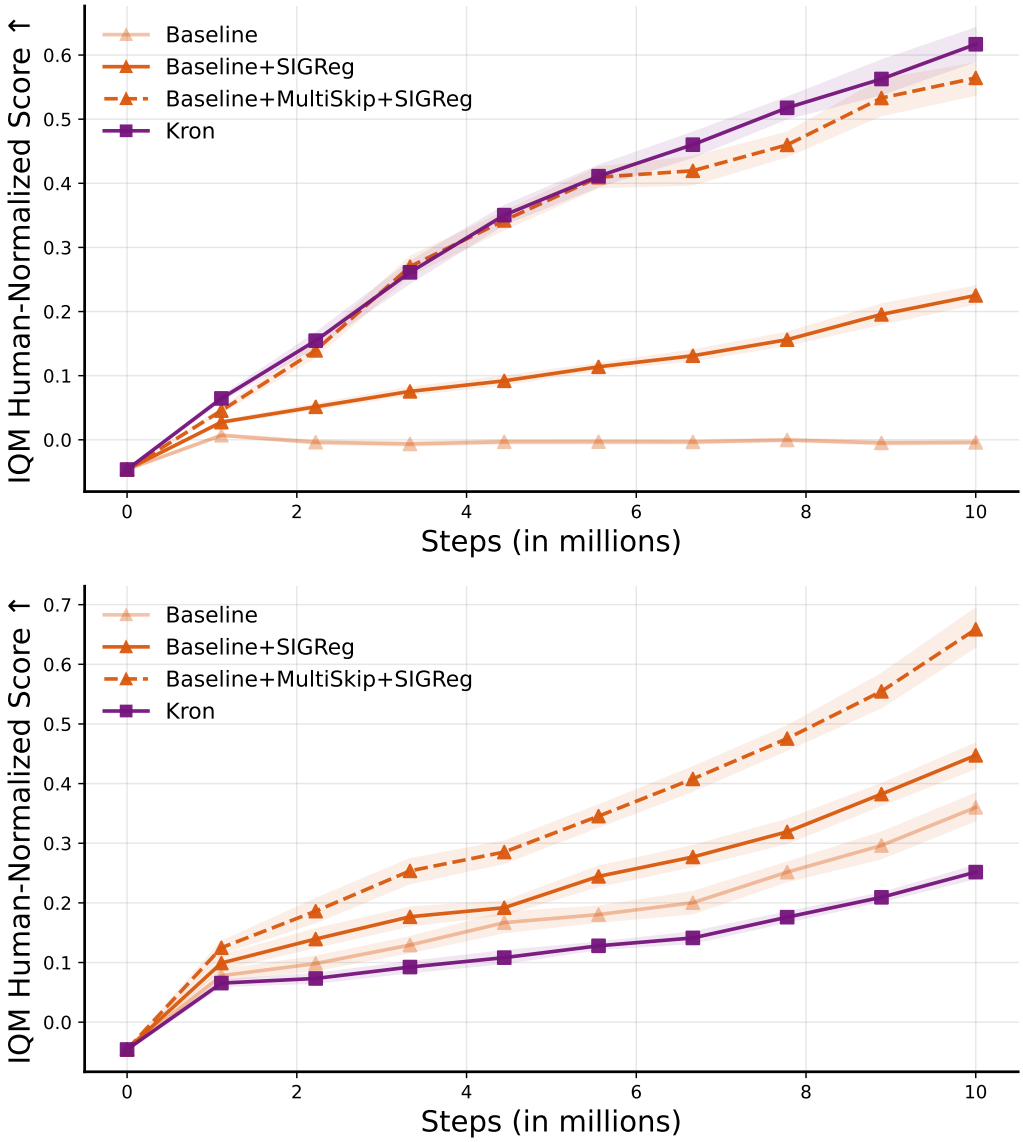

Figure 14: **IQM Human-Normalized Results for PQN (top) and PPO (bottom) Across the Atari-10 Benchmark Games.** Explicit SIGReg loss minimization encourages isotropic Gaussian embeddings, reducing the performance gap between first-order optimizers (RAdam) and second-order Kronecker-factored optimization. Adding multi-skip residual connections further improves performance.

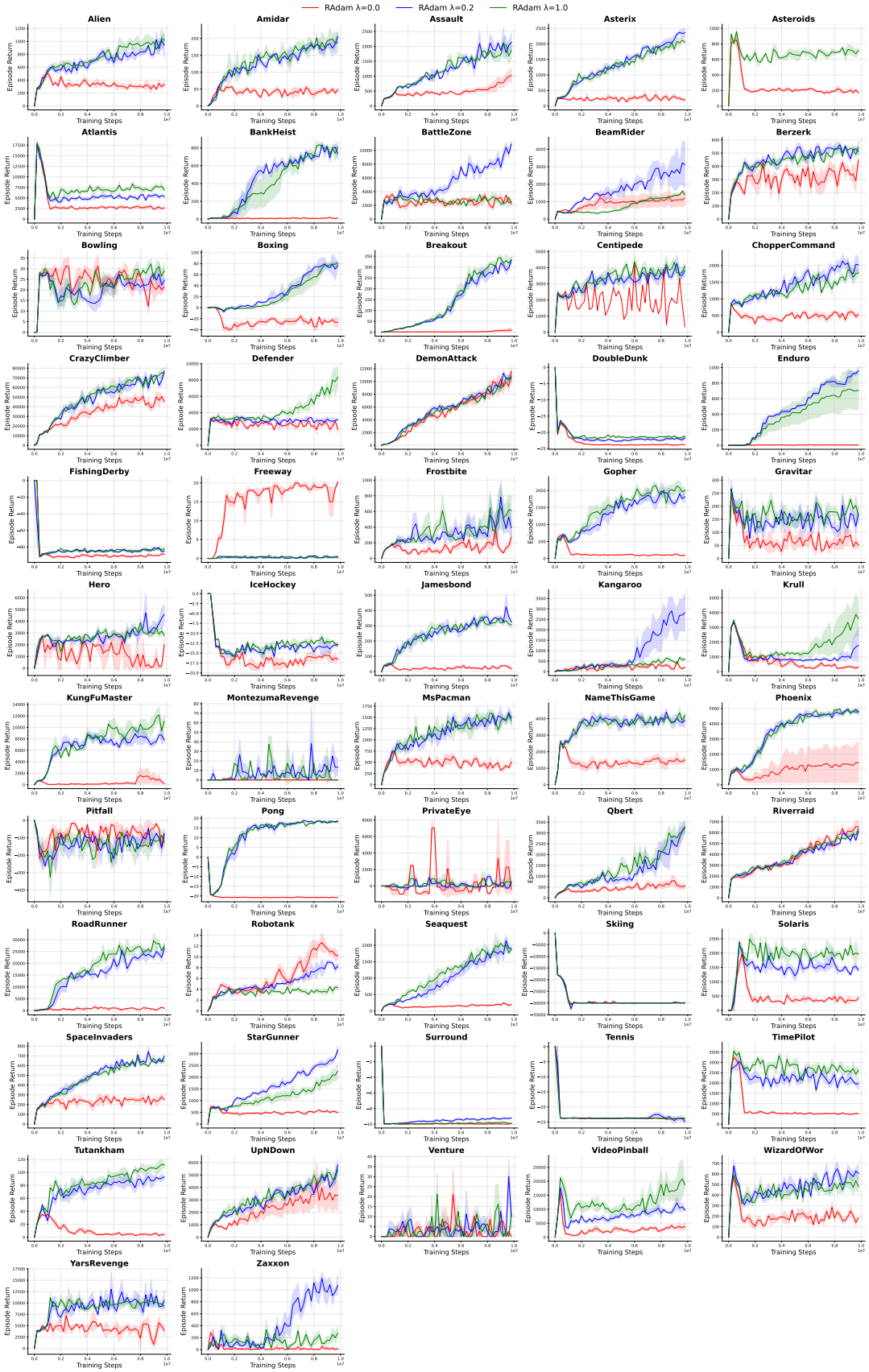

Figure 15: **PQN.** Reward over time for individual games with and without SIGReg loss minimization.

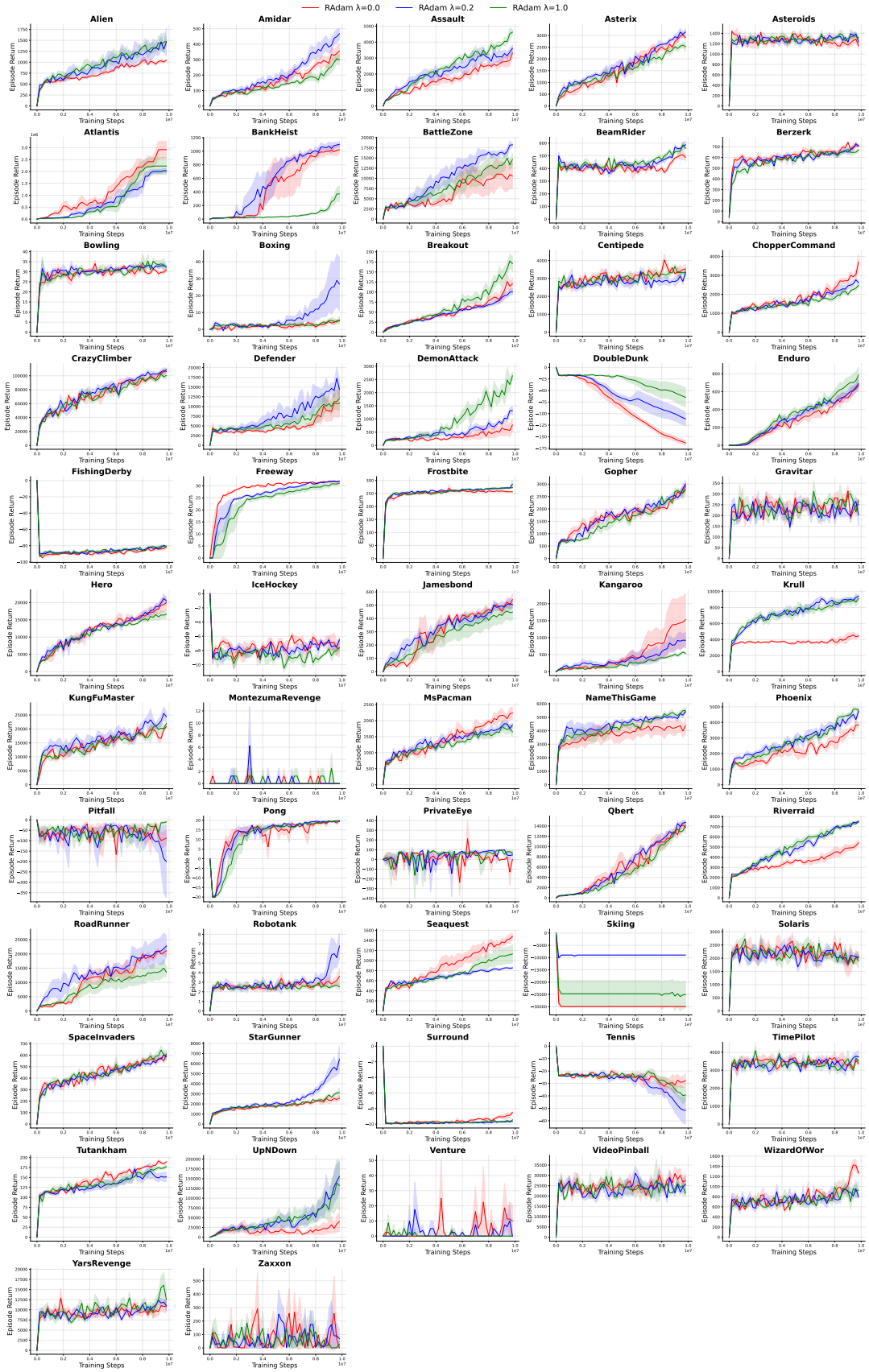

Figure 16: **PPO.** Reward over time for individual games with and without SIGReg loss minimization.

D. Isaac Gym

Across Isaac Gym continuous-control tasks (Makoviychuk et al., 2021b), enforcing isotropic Gaussian representations consistently improves learning dynamics and final performance. Compared to PPO, PPO + SIGReg achieves faster early learning, more stable training trajectories, and higher asymptotic returns across environments.

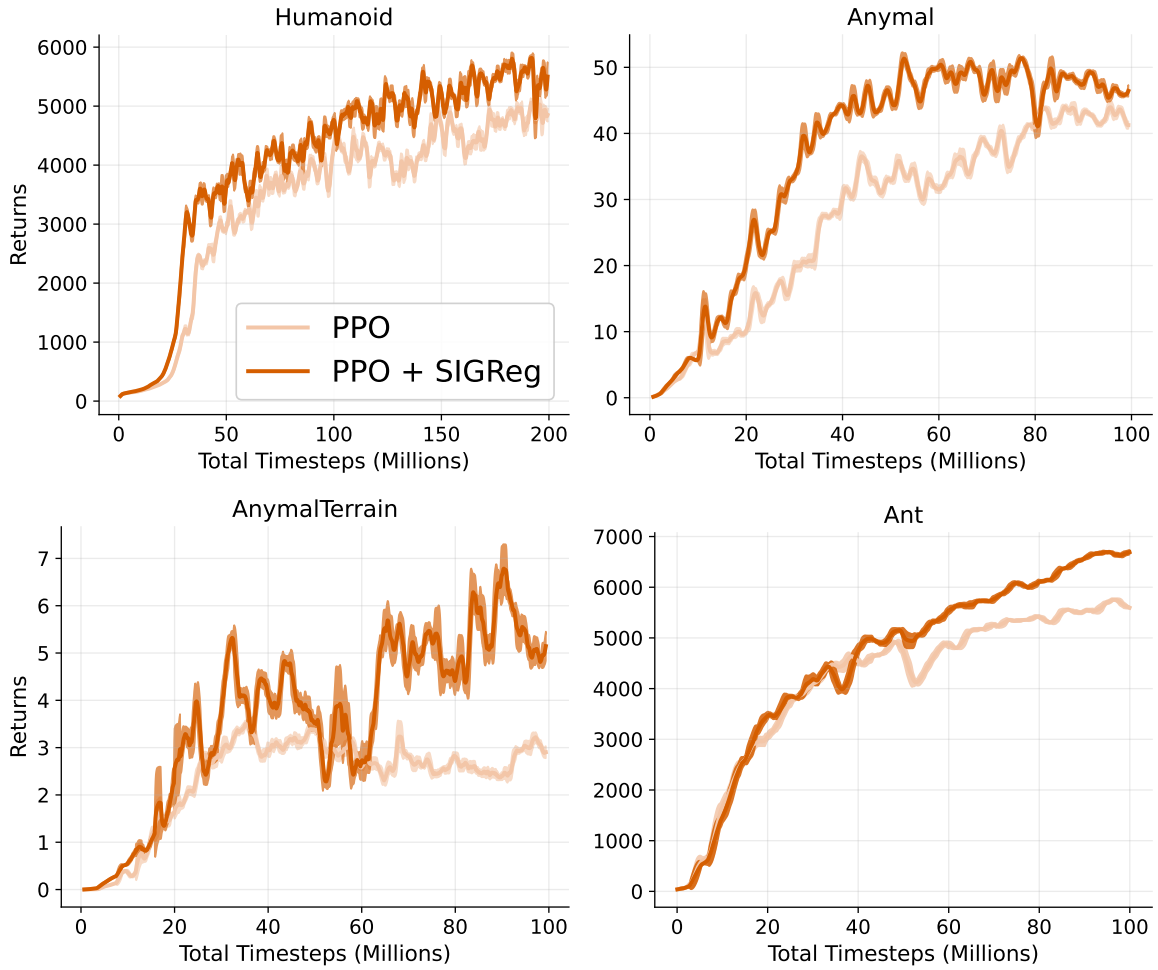

Figure 17: **Isaac Gym continuous control.** Learning curves for PPO and PPO + SIGReg on four representative locomotion tasks. Encouraging isotropic Gaussian representations improves learning stability and asymptotic performance across all environments. Curves show mean episode returns over five independent runs, with shaded regions indicating variability across runs.

E. Hyperparameters

This section summarizes the hyperparameters used across all experiments and algorithms. Unless stated otherwise, we follow the configurations proposed in the corresponding original works and adopt the same settings used in Castanyer et al. (2025) for overlapping baselines.

For consistency and computational practicality, we use a single fixed set of hyperparameters for each baseline across all environments and experimental conditions. This choice isolates the effect of the proposed methods from confounding factors introduced by per-task tuning and ensures a fair comparison across algorithms. We note, however, that deep RL methods can be sensitive to hyperparameter choices (Ceron et al., 2024a). While performing an extensive hyperparameter search for each setting could potentially yield stronger absolute performance, such a procedure is computationally prohibitive at the scale considered in this work. Importantly, our conclusions focus on relative performance trends and representation behavior, which we found to be stable under reasonable variations of the default hyperparameters.

Table 4: PQN Hyperparameters

Hyperparameter	Value / Description
Learning rate	2.5e-4
Anneal lr	False (no learning rate annealing)
Num envs	128 (parallel environments)
Num steps	32 (steps per rollout per environment)
Gamma	0.99 (discount factor)
Num minibatches	32
Update epochs	2 (policy update epochs)
Max grad norm	10.0 (gradient clipping)
Start e	1.0 (initial exploration rate)
End e	0.005 (final exploration rate)
Exploration fraction	0.10 (exploration annealing fraction)
Q lambda	0.65 (Q(λ) parameter)
Use ln	True (use layer normalization)
Activation fn	relu (activation function)

Table 5: PPO Hyperparameters

Hyperparameter	Value / Description
Learning rate	2.5e-4
Num envs	8
Num steps	128 (steps per rollout per environment)
Anneal lr	True (learning rate annealing enabled)
Gamma	0.99 (discount factor)
Gae lambda	0.95 (GAE parameter)
Num minibatches	4
Update epochs	4
Norm adv	True (normalize advantages)
Clip coef	0.1 (PPO clipping coefficient)
Clip vloss	True (clip value loss)
Ent coef	0.01 (entropy regularization coefficient)
Vf coef	0.5 (value function loss coefficient)
Max grad norm	0.5 (gradient clipping threshold)
Use ln	False (no layer normalization)
Activation fn	relu (activation function)
Shared cnn	True (shared CNN between policy and value networks)

Table 6: PPO Hyperparameters for IsaacGym

Hyperparameter	Value / Description
Total timesteps	30,000,000
Learning rate	0.0026
Num envs	4096 (parallel environments)
Num steps	16 (steps per rollout)
Anneal lr	False (disable learning rate annealing)
Gamma	0.99 (discount factor)
Gae lambda	0.95 (GAE lambda)
Num minibatches	2
Update epochs	4 (update epochs per PPO iteration)
Norm adv	True (normalize advantages)
Clip coef	0.2 (policy clipping coefficient)
Clip vloss	False (disable value function clipping)
Ent coef	0.0 (entropy coefficient)
Vf coef	2.0 (value function loss coefficient)
Max grad norm	1.0 (max gradient norm)
Use ln	False (no layer normalization)
Activation fn	relu (activation function)

Table 7: Image Classification Hyperparameters (CIFAR-10)

Hyperparameter	Value
Batch size	256
Epochs	100
Learning rate	0.00025

Table 8: SIGReg Loss Hyperparameters.

Hyperparameter	Value
Number of Random Projections	16
Number of Frequency Samples	8
Maximum Frequency	5.0
Regularization Factor (larger $\rightarrow$ stronger)	Best value between [1.0 and 0.2] (see Table 9)

Table 9: Best regularization factor (λ) selected via AUC for PQN and PPO algorithms across Atari games (Larger means stronger regularization)

Game	PQN	PPO	Game	PQN	PPO
Alien	1.0	1.0	Kangaroo	0.2	0.2
Amidar	1.0	0.2	Krull	1.0	0.2
Assault	1.0	1.0	KungFuMaster	1.0	0.2
Asterix	0.2	0.2	MontezumaRevenge	0.2	1.0
Asteroids	1.0	1.0	MsPacman	1.0	0.2
Atlantis	1.0	1.0	NameThisGame	0.2	1.0
BankHeist	0.2	0.2	Phoenix	0.2	0.2
BattleZone	0.2	1.0	Pitfall	0.2	1.0
BeamRider	0.2	1.0	Pong	0.2	0.2
Berzerk	0.2	0.2	PrivateEye	1.0	1.0
Bowling	1.0	0.2	Qbert	1.0	0.2
Boxing	0.2	0.2	Riverraid	1.0	0.2
Breakout	1.0	1.0	RoadRunner	1.0	0.2
Centipede	1.0	1.0	Robotank	0.2	0.2
ChopperCommand	0.2	0.2	Seaquest	1.0	1.0
CrazyClimber	1.0	0.2	Skiing	0.2	0.2
Defender	1.0	0.2	Solaris	1.0	1.0
DemonAttack	0.2	1.0	SpaceInvaders	0.2	1.0
DoubleDunk	1.0	1.0	StarGunner	0.2	0.2
Enduro	0.2	1.0	Surround	0.2	1.0
FishingDerby	1.0	1.0	Tennis	0.2	1.0
Freeway	1.0	0.2	TimePilot	1.0	1.0
Frostbite	1.0	0.2	Tutankham	1.0	1.0
Gopher	1.0	0.2	UpNDown	1.0	1.0
Gravitar	1.0	1.0	Venture	0.2	0.2
Hero	1.0	0.2	VideoPinball	1.0	1.0
IceHockey	1.0	0.2	WizardOfWor	0.2	0.2
Jamesbond	0.2	0.2	YarsRevenge	1.0	1.0
			Zaxxon	0.2	0.2